HGTFN: A Hypergraph and Transformer Fusion Network for Hyperspectral Image Classification

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***Abstract*—Graph neural networks (GNNs) have demonstrated significant potential in hyperspectral image(HSI) classification. However, their performance is often constrained by fixed graph structures and limited capacity for long-range dependency modeling. To address these limitations, we propose a novel hypergraph and transformer fusion network (HGTFN) that integrates hypergraph convolution and transformer encoding in a complementary manner, combining localized structural modeling with global semantic context aggregation. Specifically, a adaptive multihop hypergraph construction (SMHGC) module is developed to generate adaptive, multiscale hypergraphs guided by spatial–spectral correlations, enabling the capture of higher order relational structures. This is followed by a hypergraph convolution with edge-wise attention (HGCEA) module, which enhances feature propagation through edge-adaptive weighting. To overcome the rigidity of grid-based tokenization in conventional vision transformers (ViTs), we introduce a key-pixel guided transformer (KPFormer), which constructs localized neighborhood tokens centered on key pixels based on feature similarity and employs a context-aware feedforward block to strengthen local–global interactions. Furthermore, a residual complement fusion network (RCFN) is incorporated to fuse multisource features and refine semantic representation in nonkey regions. This design enables HGTFN to generate more balanced and discriminative representations even with limited training samples. Experiments conducted on four benchmark datasets, namely Indian Pines (IPs), Pavia University (PU), Salinas (SA), and WHU-Hi-HongHu (HongHu), demonstrate that the HGTFN model achieves state-of-the-art performance with overall accuracies (OAs) of 95.73%, 95.58%, 99.19%, and 96.56%, respectively, significantly outperforming ten other algorithms, including convolutional neural network (CNN), transformer, and graph-based methods. The core code for HGTFN is published at https://github.com/Majunyi310321/HGTFN**

***Index Terms*—Convolutional network, graph neural network (GNN), hyperspectral image classification (HSIC), long-range dependency modeling, transformer.**

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I. Introduction

HYPERSPECTRAL image (HSI), characterized by its nanometer-level spectral resolution and continuous band acquisition, enables the simultaneous capture of both spatial and spectral information across hundreds of narrow and contiguous bands [1]. This unique capability allows for joint spatial–spectral analysis, providing discriminative information for precise material identification. Due to its high-dimensional spectral representation, HSI has been extensively utilized in hyperspectral image classification (HSIC), playing a pivotal role in a wide range of remote sensing applications such as precision agriculture, mineral exploration, environmental monitoring, and urban land-cover classification [2], [3], [4], [5].

Despite the significant progress achieved in HSIC, several challenges remain unresolved. First, HSI data are typically characterized by high dimensionality [6], limited labeled samples [7], and severe class imbalance [8], which often lead to model overfitting [9]. Second, hyperspectral bands contain substantial redundancy and strong interband correlations, increasing the difficulty of effective feature extraction and representation. Moreover, in large-scale remote sensing scenes, complex object structures and blurred class boundaries further complicate the classification task [10]. Accurately modeling both fine-grained local structures and global contextual dependencies remains a critical challenge for improving classification accuracy [11].

Early HSIC methods mainly relied on traditional machine learning algorithms such as decision trees, random forests, K-nearest neighbors (KNNs), and support vector machines (SVMs) [12], [13]. While these methods are structurally simple, they primarily exploit spectral features without effectively utilizing spatial context [14], and often require extensive manual feature engineering and parameter tuning [15].

Deep learning-based HSIC methods enable automatic hierarchical feature extraction through data-driven learning. Among the various architectures, such as recurrent neural networks (RNNs), autoencoders, and convolutional neural networks (CNNs), models based on CNNs are the most popular.

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He et al. [16] introduced a multiscale 3-D-CNN for joint spectral–spatial feature learning. Zhong et al. [17] enhanced this with residual connections, forming the spectral–spatial residual network (SSRN) model. Roy et al. [18] further proposed HybridSN, combining 3-D-CNNs and 2-D-CNNs for multilevel feature extraction. To better capture spectral–spatial dependencies, Zhu et al. [19] designed RSSAN, incorporating dual attention to adaptively focus on informative bands and pixels.

However, due to the high dimensionality and strong interband correlations inherent in HSI data, conventional CNNs—with their fixed local receptive fields—remain limited in modeling long-range spatial dependencies. Furthermore, the progressive downsampling via pooling layers inevitably leads to the loss of fine-grained spatial details [20], [21].

To overcome the inherent limitations of CNNs, a novel architecture known as the transformer was introduced in [22] for modeling sequential data. By computing pairwise attention weights, the transformer establishes direct interactions between any two positions in a sequence, enabling the capture of long-range dependencies. Inspired by this mechanism, Alexey [23] extended the transformer to computer vision tasks, where images were divided into patches and processed using a pure transformer framework, resulting in the vision transformer (ViT). In the hyperspectral domain, Hong et al. [24] developed SpectralFormer (SF), which employed group-wise spectral embedding and cross-layer adaptive fusion (CAF) to achieve effective spectral–spatial feature fusion. Zhang et al. [25] proposed a lightweight ViT variant (LiT), where a controlled multiclass stratified sampling strategy was used to generate properly sized patch inputs, reducing overfitting by minimizing overlap between training and testing regions. Furthermore, Ouyang et al. [26] presented HybridFormer, which integrated multigranularity labels and a spatial–spectral attention mechanism to enhance multiscale object recognition.

Despite their global modeling capability, transformer-based models often fail to preserve fine-grained boundaries in complex remote sensing scenes. At the same time, their computational complexity grows quadratically with image size, which poses significant challenges for high-dimensional HSI cubes. In addition, vanilla transformer architectures are designed for Euclidean data structures, making it difficult to capture topological relationships in irregular or non-Euclidean regions. This limits their performance in complex and heterogeneous landscapes.

Graph CNNs (GCNs) have emerged as a powerful deep learning paradigm for processing non-Euclidean data structures, making them well-suited for HSIC. By representing the relationships between pixels or superpixels as nodes and edges in a graph—where edges encode spatial or spectral similarity—graph neural networks (GNNs) naturally capture the topological structure of land-cover regions [21]. By applying convolutional operations on graphs, GCNs effectively capture both spectral features and spatial dependencies, thereby enhancing classification accuracy and robustness. According to their convolutional strategy, GCN-based approaches can be categorized into two classes: spectral-based and spatial-based methods [27].

A. Spectral-Based GCNs

Spectral-based graph convolution methods interpret graph filtering as a denoising process over graph signals, inspired by signal processing theory. In 2018, GCNs were first applied to HSIC, enabling low-dimensional visualization and classification of hyperspectral data [28]. In the same year, Qin et al. [29] simultaneously incorporated both spectral and spatial information into the graph convolution process, resulting in more comprehensive feature representations. To further enhance multimodal feature extraction, Ma et al. [30] integrated CNN-based spatial features with spectral and semantic features from multiple levels. While these pixel-level spectral GCN approaches are straightforward and effective, they tend to be sensitive to noise and computationally intensive when applied to large-scale hyperspectral cubes. To address these limitations, Wan et al. [31] proposed a context-aware dynamic GCN, which performs automatic pixel-to-superpixel conversion and improves classification performance in irregular or heterogeneous regions. Ding et al. [32] introduced a globally consistent graph convolutional network that constructs a global graph structure via superpixel segmentation, effectively capturing global contextual information through topological consistency. More recently, Ding et al. [33] proposed a subgraph-dependent neural network (SGDNet), which leverages spectral feature diversity and local subgraph structures to alleviate neighbor explosion and optimize node representations in spectral GCNs. Most recently, Yu et al. [34] developed a contrastive GCN framework for HSIC that leverages inherent supervisory signals in hyperspectral data to achieve robust performance, even with limited labeled samples.

B. Spatial-Based GCNs

Spatial-based graph convolutional methods focus on aggregating features from each node and its neighboring nodes in the graph structure, enabling the capture of local spatial dependencies. Compared to spectral-based methods, spatial GCNs avoid complex eigen-decomposition and are generally more interpretable and computationally efficient. In 2020, Yang et al. [35] first introduced GraphSAGE to HSIC, leveraging a neighborhood sampling and aggregation mechanism that significantly reduces spatial complexity. In the same year, Sha et al. [36] incorporated the graph attention network (GAT) into semi-supervised HSI classification, effectively addressing the challenges posed by high-dimensionality and limited labeled samples. To further enhance label prediction under few-shot settings, Zayats and Ostendorf [37] proposed a novel edge-labeled GNN, where node and edge features are iteratively updated to obtain more accurate edge labels. Ding et al. [38] introduced LPDGCN, a spatial GCN that exploits the positional distributions of labeled superpixels to enhance classification robustness, particularly in boundary-sensitive or sparsely labeled regions. More recently, Shi et al. [39] introduced a key sample selection strategy that identifies hard-to-distinguish unlabeled samples and incorporates them into training. This method significantly improves the model's discriminative capability and yields superior classification performance even with scarce labeled data.

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Although GNNs have been widely applied in HSIC, most methods rely on static KNN-based graphs, which struggle to capture fine-grained local structures and higher order relationships. To address this, Duan et al. [40] proposed a structure-preserving hypergraph convolutional network (HGCN) to aggregate discriminative features across regions, but it depends on fixed neighborhood criteria. Wang et al. [41] further introduced HGCN-MHF, which replaces KNN with region-based topology to construct more adaptive hyperedges. To support spectral modeling, Chan et al. [42] developed a hypergraph Laplacian operator, laying the foundation for spectral HGCNs. However, spectral methods suffer from high complexity and poor scalability. F^2HNN [30] introduces a feature fusion hypergraph network that integrates multimodal spectral–spatial features via hyperedge construction, offering improved discriminability in complex scenes. Spatial-domain HGCNs, such as HGNN+ [43], offer more flexibility by learning modality-specific hyperedge weights, but still treat all nodes equally within each hyperedge, reducing feature discriminability. In addition, a subspace clustering method based on hypergraph convolution, termed HGCSC, was proposed by Zhang et al. [44]. By modeling high-order relationships and long-range dependencies in hyperspectral data, HGCSC significantly improves clustering accuracy. Furthermore, an Enhanced hypergraph neural network (HGNN) was introduced by Wang et al. [45], which incorporates a key hypergraph structure and establishes an end-to-end CNN–HGNN framework. This design effectively addresses the limitations of conventional HGCNs in HSIC, particularly the insufficient semantic representation and loss of pixel-level details. More recently, CHGUnet [46] proposes a clustering-based hypergraph U-Net architecture with high-order topology-aware pooling and unpooling, enhancing structural feature learning.

Despite the recent advancements, existing hypergraph-based HSIC methods still encounter several significant limitations. First, most HGCNs rely on static or manually constructed hypergraphs, which are either sensitive to noise or incapable of flexibly capturing diverse structural patterns in heterogeneous scenes. Second, conventional spatial-domain HGCNs typically assign equal importance to all vertices within a hyperedge, disregarding the varying contributions of individual nodes. This uniform treatment undermines semantic consistency and robustness, particularly in mixed-category or boundary regions. Third, hypergraph convolutions inherently operate on local neighborhoods (i.e., hyperedges) and thus lack the capacity to explicitly model long-range global dependencies across the entire image.

To address these challenges and simultaneously leverage both the spatial–spectral characteristics of hyperspectral data and the complementary strengths of HGCNs and transformers, we propose a novel hybrid spatial–spectral classification framework named hypergraph and transformer fusion network (HGTFN), as illustrated in Fig. 1. Specifically, an adaptive multihop hypergraph construction (SMHGC) module is developed to adaptively integrate spectral similarity and spatial topology into a unified hypergraph structure. To further distinguish the contributions of nodes within each hyperedge, an Edge-wise Attention mechanism is incorporated, resulting in the proposed hypergraph convolution with edge-wise attention (HGCEA). In addition, a key-pixel selection block is designed to extract the most informative tokens for transformer inference. Combined with a localized similarity-aware tokenization strategy, the proposed design enables computational efficiency while enhancing the discriminative representation of key regions. In summary, the primary contributions of this article are as follows.

1) To harness the complementary strengths of GCNs and transformers, a hybrid architecture is proposed by integrating the HGCN with edge-wise attention and the key-pixel guided transformer (KPFormer). This hybrid parallel and serial design enables the joint exploitation of global pixel-level context and local superpixel-level cues for enhanced feature fusion.
2) A superpixel-based multihop hypergraph construction module is developed to address the expressiveness limitations of kNN graphs. It adaptively integrates three types of hyperedges—local adjacency, mid-range structural, and feature-aware semantic—to enhance hypergraph robustness and representation.
3) An edge-wise attention mechanism is incorporated into hypergraph convolution to account for node-wise contribution disparities within each hyperedge, thereby improving semantic consistency and suppressing noise propagation.
4) The KPFormer module is introduced to overcome the rigidity of grid-based patch embeddings in conventional ViTs. By performing key-pixel selection and constructing localized neighborhood tokens based on structural similarity, KPFormer generates more discriminative representations for fine-grained classification. Moreover, a context-aware feedforward block enhances local–global feature fusion while maintaining computational efficiency.
5) A residual complement fusion network (RCFN) that fuses features from both key and nonkey pixels via a residual pathway, enabling dense per-pixel predictions with global consistency and structural sensitivity.

The structure of this article is arranged as follows. Section II reviews related work. Section III provides a comprehensive explanation of the proposed algorithm. Experimental settings and results are reported in Section IV, with further discussion and analysis presented in Section V. Finally, Section VI offers the conclusion of this study.

II. Related Work

In this section, we provide a brief overview of two representative models commonly used in HSI classification: GNNs and transformers. The review focuses on their fundamental principles.

A. Hypergraph Neural Network

HGNNs extend conventional GNNs by modeling high-order relationships among multiple nodes through the use of hyperedges. Unlike standard graphs, where an edge connects only two nodes, a hypergraph allows each hyperedge to connect

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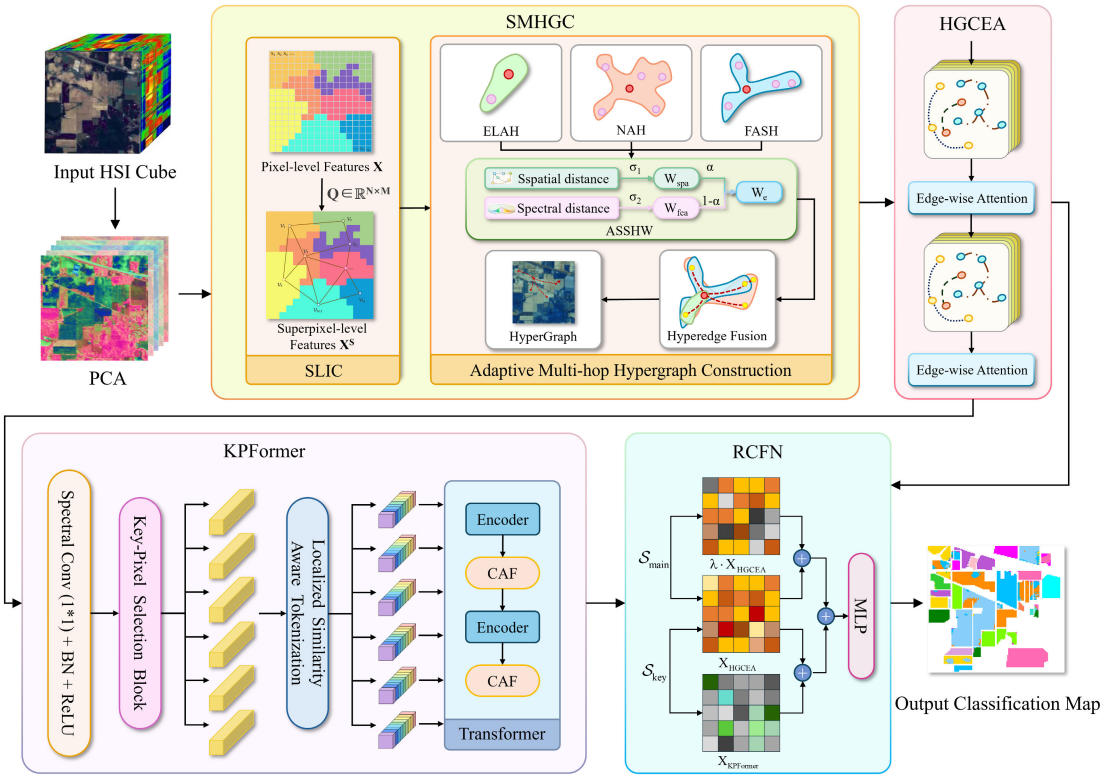


Fig. 1. Overall architecture of HGTFN. The input HSI is first processed by PCA and SLIC to obtain pixel- and superpixel-level features. The SMHGC module builds adaptive multihop hypergraphs using fused spatial–spectral hyperedges. These hypergraphs are passed into the HGCEA module to extract structural features, which are subsequently forwarded to KPFormer for key-pixel semantic modeling—forming a serial path. Finally, the outputs from both HGCEA and KPFormer are fused in parallel by the RCFN module to generate the final classification map.

an arbitrary number of nodes, enabling the representation of more complex and structured interactions within the data.

Formally, a hypergraph is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_1, v_2, \ldots, v_i\}$ denotes the set of N vertices and $\mathcal{E} = \{e_1, e_2, \ldots, e_t\}$ denotes the set of E hyperedges. The relationships between vertices and hyperedges are typically encoded using an incidence matrix $\mathbf{H} \in \mathbb{R}^{N \times M}$

$$\mathbf{H}(v, e) = \begin{cases} 1, & \text{if } v_i \in e_t \\ 0, & \text{if } v_i \notin e_t \end{cases} \tag{1}$$

where $v_i \in e_t$ denotes a vertex v_i that is connected by the hyperedge e_t. Based on these definitions, the vertex degree matrix $\mathbf{D}_v$ and the hyperedge degree matrix $\mathbf{D}_e$ can be computed as follows:

$$\mathbf{D}_v = \text{diag}\left(\sum_{t=1}^{E} \mathbf{H}(i,t)\right), \quad i = 1, 2, \ldots, M \tag{2}$$

$$\mathbf{D}_e = \text{diag}\left(\sum_{i=1}^{M} \mathbf{H}(i,t)\right), \quad t = 1, 2, \ldots, E \tag{3}$$

where diag$(\cdot)$ denotes the operation that converts a vector into a diagonal matrix. To perform convolution operations on hypergraphs, most HGNNs adopt either spectral or spatial approaches. In spectral HGNNs, hypergraph Laplacians are generalized from traditional graph Laplacians to enable signal filtering in the spectral domain. The conventional multilayer hypergraph convolution can then be formulated as

$$\mathbf{X}^{(l+1)} = \sigma\left(\mathbf{D}_v^{-\frac{1}{2}}\mathbf{H}\mathbf{W}\mathbf{D}_e^{-1}\mathbf{H}^{\top}\mathbf{D}_v^{-\frac{1}{2}}\mathbf{X}^{(l)}\Theta^{(l)}\right) \tag{4}$$

where $\mathbf{W}$ is a diagonal matrix containing the weights of all hyperedges. $\mathbf{D}_v$ and $\mathbf{D}_e$ denote the degree matrices of the nodes and hyperedges, respectively. $\sigma(\cdot)$ represents a nonlinear activation function, such as ReLU. This formulation propagates features from each vertex to its associated hyperedges and then back to the connected vertices, weighted by hyperedge importance and normalized by vertex and hyperedge degrees.

Due to their ability to capture complex many-to-many relationships, HGNNs have demonstrated strong potential in various tasks involving irregular or structured data, such as point cloud processing, social network analysis, and particularly HSIC.

B. Transformer

The transformer architecture addresses the drawbacks of conventional RNNs, which rely on sequential computation, and CNNs, which struggle to capture long-range dependencies. This is accomplished by employing a self-attention mechanism that allows global context modeling across all positions in the input sequence in a fully parallel and order-independent fashion. The standard transformer is composed of an encoder–decoder structure; however, for image classification tasks, only the encoder is utilized, as shown in Fig. 2.

Given an input sequence $\mathbf{X} = x_1, x_2, \ldots, x_p$ of length p, where each $x_p \in \mathbb{R}^d$ represents a d-dimensional feature vector, the encoder transforms $\mathbf{X}$ into a continuous latent representation $\mathbf{Z} = z_1, z_2, \ldots, z_p$. In the original model, a decoder autoregressively generates the output sequence, but this component is excluded in our classification scenario.

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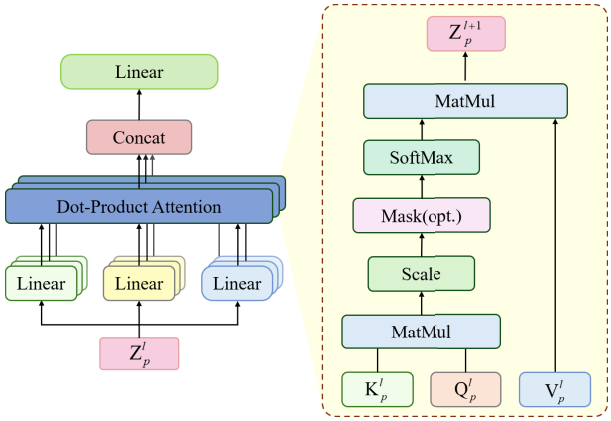


Fig. 2. Architecture of the proposed key-pixel transformer. Optional masking operations can be applied to restrict the attention scope.

To embed positional information, each token x_i is combined with a positional encoding vector p_i to form x_i'

$$x_i' = x_i + p_i. \tag{5}$$

In the self-attention block, each x_i' is linearly projected into three representations: query (Q), key (K), and value (V). Denoting $\mathbf{X}' = [x_1', x_2', \ldots, x_p']^\top$, the transformation is formulated as

$$Q = \mathbf{X}'W^Q, \quad K = \mathbf{X}'W^K, \quad V = \mathbf{X}'W^V. \tag{6}$$

The self-attention output is then computed using the scaled dot-product mechanism

$$\mathbf{Z} = \text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_h}}\right)V \tag{7}$$

where d_h denotes the dimensionality of queries and keys. In the case of multihead self-attention, the network concurrently attends to multiple representation subspaces, allowing it to capture diverse semantic dependencies across positions.

III. Proposed Methodology

A. Superpixel-Based Multihop Hypergraph Construction

In HSI classification, conventional GNNs suffer from limited structural representation capability, as their message propagation is typically constrained to second-order neighborhood connections, making them prone to noise accumulation. Most existing HGCNs rely on fixed KNN strategies for hyperedge construction, which leads to insufficient structural diversity and heavy reliance on handcrafted hyperedge weighting and composition schemes. To enhance graph modeling expressiveness, recent work [41] has introduced multihyperedge fusion strategies that integrate multiple hyperedge types to better capture topological structures. However, these approaches still depend on static or manually defined hypergraphs, making them sensitive to noise.

To address these limitations, we propose a superpixel-based multihop hypergraph construction module that combines multihop and multitype hyperedges, significantly improving the model's capacity to capture complex spatial–spectral dependencies. On this basis, we introduce an adaptive spatial–spectral hyperedge weighting (ASSHW) strategy to improve hypergraph construction for HSI tasks. This strategy first models multiscale structural relations via three complementary hyperedge types: local adjacency, mid-range topology, and spectral similarity. It then employs a learnable weighting mechanism that integrates both spatial and spectral information into hyperedge representations, enabling dynamic selection and enhancement based on data content. Unlike traditional methods that rely on fixed parameters, ASSHW allows the network to adaptively learn the importance of different hyperedges in an end-to-end manner, thereby significantly improving the generalization of structural modeling.

Specifically, for the HSI $\mathbf{X} \in \mathbb{R}^{H\times W\times d}$ (after PCA dimensionality reduction), SLIC is adopted to segment the image into M superpixel regions. To facilitate subsequent processing, $\mathbf{X}$ is spatially flattened into a pixel-wise matrix as

$$\tilde{\mathbf{X}} = \text{Flatten}(\mathbf{X}), \qquad \tilde{\mathbf{X}} \in \mathbb{R}^{HW\times d} \tag{8}$$

where each row $\tilde{\mathbf{X}}_i$ corresponds to the spectral feature vector of the ith pixel in the image.

Let $\Omega = \{1, 2, \ldots, HW\}$ denote the index set of all pixels, and let $\{\mathcal{V}_1, \ldots, \mathcal{V}_M\}$ denote the obtained superpixel partition, where $\mathcal{V}_j$ contains the indices of pixels belonging to the jth superpixel. The pixel–superpixel assignment matrix $\mathbf{Q} \in \mathbb{R}^{HW\times M}$ is then defined as

$$\mathbf{Q}_{i,j} = \begin{cases} 1, & \text{if } i \in \mathcal{V}_j \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

Based on the assignment matrix $\mathbf{Q}$, superpixel-level features are obtained by aggregating the pixel features within each superpixel. Specifically, we adopt average pooling to compute the superpixel representation

$$\mathbf{X}_S = \text{Encoder}(\tilde{\mathbf{X}}; \mathbf{Q}) = \mathbf{D}_s^{-1}\mathbf{Q}^\top\tilde{\mathbf{X}} \in \mathbb{R}^{M\times d} \tag{10}$$

where $\mathbf{D}_s = \text{diag}(\mathbf{Q}^\top\mathbf{1})$ is the diagonal degree matrix of superpixels, and $\mathbf{1} \in \mathbb{R}^{HW}$ denotes an all-ones vector. The superpixel features are then projected back to the pixel space through a decoding operation

$$\mathbf{X}_E = \text{Decoder}(\mathbf{X}_S; \mathbf{Q}) = \mathbf{Q}\mathbf{X}_S \in \mathbb{R}^{HW\times d} \tag{11}$$

where each pixel inherits the feature representation of its corresponding superpixel.

To better model the physical structures within hyperspectral scenes using geometric information, an adjacency matrix $A \in \mathbb{R}^{M\times M}$ is employed to encode the connectivity among superpixel nodes

$$\mathbf{A}_{ij} = \begin{cases} 1, & \text{if } v_i \text{ and } v_j \text{ are adjacent} \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

Based on the initial superpixel segmentation, a multistrategy hypergraph construction framework is proposed to enhance the structural representation capability of the hypergraph from both spatial and spectral perspectives. As shown in the SMHGC block in Fig. 1, each superpixel node is sequentially treated as a center node, and three complementary types of hyperedges are constructed to capture structural dependencies at different spatial scales. This results in three distinct groups

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of hyperedges. The first type is referred to as edge-level local adjacency hyperedges (ELAHs). These hyperedges are constructed from pairwise spatial connections between adjacent superpixels derived from the SLIC segmentation. ELAH's focus on fine-grained boundary modeling and preserving the local geometric structure of the scene, thereby enhancing edge-level discriminative capability

$$e_{\mathrm{ELAH}} = \left\{ \{v_i, v_j\} \;\middle|\; \mathbf{A}_{ij} = 1, i < j \right\}. \tag{13}$$

The second type is the neighborhood aggregation hyperedges (NAHs), which are constructed based on the first-order spatial neighborhood (1-hop) of each superpixel node. This neighborhood captures the local contextual structure of the region, enhancing spatial consistency and mitigating fragmentation in spatially continuous targets. The NAH formulation is defined as follows:

$$e_{\mathrm{NAH}} = \left\{ \{v_i, v_j, v_k\} \;\middle|\; \mathbf{A}_{ij} = 1, \mathbf{A}_{jk} = 1, v_i \neq v_k \right\}. \tag{14}$$

The third type is the feature-aware semantic hyperedges (FASHs), which are constructed within the second-order spatial neighborhood (2-hop) by selecting the top K most spectrally similar superpixels. These nonlocal hyperedges enable long-range aggregation of spectrally consistent but spatially distant regions, thereby enhancing global semantic consistency. The construction is formulated as follows:

$$e_{\mathrm{FASH}} = \left\{ \{v_i\} \cup \mathcal{N}_K^{\mathrm{local}}(i) \;\middle|\; i = 1, 2, \ldots, M \right\} \tag{15}$$

where $\mathcal{N}_K^{\mathrm{local}}(i)$ denotes the set of the K_{top} nearest superpixels to node i in the spectral feature space, selected from its 2-hop spatial neighbors $\mathcal{N}^{(\text{2-hop})}(i)$. This diversified hyperedge generation strategy facilitates the integration of rich spatial and spectral information, thereby significantly enhancing the representational capacity of the hypergraph. Each final hyperedge is constructed by concatenating and fusing the three aforementioned types of hyperedges. The complete hyperedge set is defined as follows:

$$e = e_{\mathrm{ELAH}} \cup e_{\mathrm{NAH}} \cup e_{\mathrm{FASH}}. \tag{16}$$

Although the proposed hyperedges are generated from three different strategies, they are designed to capture complementary structural relations at distinct spatial and semantic scales. To further improve the discriminative capability of the hypergraph, the proposed ASSHW strategy adaptively reweights different hyperedges based on their spatial compactness and spectral similarity, enabling effective integration of complementary structural information. Specifically, a hyperedge can connect multiple nodes, and for the tth hyperedge e_t, we uniformly compute the spatial distance $D_{spa}(e_t)$ and the spectral distance $D_{fea}(e_t)$ between all the nodes inside it, which is expressed as follows:

$$D_{\mathrm{spa}}(e_t) = \frac{1}{K_t(K_t - 1)} \sum_{i \neq j} \left\| \mathbf{p}_i - \mathbf{p}_j \right\|_2 \tag{17}$$

$$D_{\mathrm{fea}}(e_t) = \frac{1}{K_t(K_t - 1)} \sum_{i \neq j} \left\| \mathbf{s}_i - \mathbf{s}_j \right\|_2 \tag{18}$$

where K_t denotes the number of nodes within the tth hyperedge. And $\|\cdot\|_2$ represents the Euclidean norm. For each superpixel v_i, $\mathbf{p}_i$ denotes its centroid coordinate in the spatial domain, and $\mathbf{s}_i$ represents its spectral feature vector. Furthermore, the spatial measurement weight $W_{\mathrm{spa}}(e)$ and the spectral measurement weight $W_{\mathrm{fea}}(e)$ of e_t can be obtained

$$W_{\mathrm{spa}}(e_t) = \exp\left(-\frac{D_{\mathrm{spa}}(e_t)}{2\sigma_1^2} \right) \tag{19}$$

$$W_{\mathrm{fea}}(e_t) = \exp\left(-\frac{D_{\mathrm{fea}}(e_t)}{2\sigma_2^2} \right) \tag{20}$$

where σ_1 and σ_2 are sensitivity parameters that regulate the influence of spatial and spectral distances on the hyperedge weights. To enable adaptive weighting, σ_1 and σ_2 are computed from the global distance matrices, allowing the Gaussian kernel to dynamically adjust to the structural characteristics of different datasets. Finally, a differentiable fusion strategy is employed to integrate the spatial and spectral weights into a unified representation:

$$\alpha = \frac{\sigma_1}{\sigma_1 + \sigma_2} \tag{21}$$

$$W_e = \alpha \cdot W_{\mathrm{spa}}(e_t) + (1 - \alpha) \cdot W_{\mathrm{fea}}(e_t) \tag{22}$$

where α is a tunable parameter that enables adaptive modulation of the fusion process across different data distributions. This linear weighting scheme allows for a smooth tradeoff between spatial compactness and spectral similarity. Moreover, the formulation is differentiable and can be optimized via gradient-based learning, providing enhanced robustness under noisy or inconsistent distance measurements.

In summary, the proposed SMHGC module achieves two key advancements over traditional graph construction methods. First, by introducing three complementary types of hyperedges—ELAH, NAH, and FASH—it enables joint modeling of fine-grained spatial boundaries, regional contextual consistency, and long-range semantic dependencies. Second, the proposed ASSHW strategy dynamically balances the importance of spatial and spectral features during end-to-end training, significantly enhancing the robustness and generalization ability of the constructed hypergraph. Overall, this module not only mitigates the reliance on handcrafted priors in conventional hypergraph construction but also provides a high-quality topological foundation for subsequent hypergraph convolution and transformer-based feature extraction.

B. HGCN With Edge-Wise Attention

HGNN+ proposed a general spatial-domain high-order multimodal hypergraph convolution framework [43]. However, this approach aggregates all nodes within a hyperedge equally, disregarding the varying importance of individual nodes, which limits the model's discriminative power. To enhance the representational capacity of node features within each hyperedge, a structure-aware convolutional module is introduced in this work by incorporating an Edge-wise Attention mechanism into the conventional hypergraph convolution. Compared to global spatial attention mechanisms, the proposed edge-wise attention significantly reduces computational complexity, while enabling finer modeling of higher order intrahyperedge relationships and contextual dependencies.

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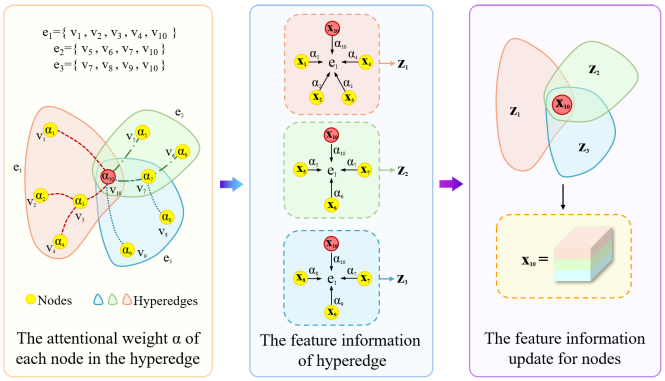


Fig. 3. Architecture of the HGCEA. Each node v_i participates in multiple hyperedges e_t, and the attention weights α are learned to capture the relative importance of nodes within each hyperedge. Feature information is first aggregated at the hyperedge level using attention-based weighting, and then fused across all hyperedges to update the target node's representation.

Building upon the conventional hypergraph structure, we propose an innovative edge-wise attention mechanism in hypergraphs, which explicitly models intrahyperedge dependencies by computing attention weights solely among nodes within each hyperedge, as shown in Fig. 3, through a vertex–hyperedge–vertex message passing scheme, this approach enables the extraction of deeper spatial contextual information in a lightweight and localized manner. Specifically, for each hyperedge ε_k, attention scores are computed among all nodes within the hyperedge to quantify their relative importance in the feature aggregation process

$$\alpha_i^{(t)} = \frac{\exp\left(\mathbf{x}_i^\top \Theta^{(l)} \cdot \mathbf{x}_i\right)}{\sum_{r\in e_t} \exp\left(\mathbf{x}_r^\top \Theta^{(l)} \cdot \mathbf{x}_r\right)} \quad \forall i \in e_t \tag{23}$$

where $\alpha_i^{(t)}$ denotes the attention weight of node i within the tth hyperedge e_t, which contains r connected nodes. $\mathbf{x}_i$ represents the feature vector of node i, and $\Theta^{(l)}$ is a learnable parameter used for attention scoring. Subsequently, the feature representation of hyperedge e_t is obtained by performing a weighted aggregation over all its constituent node features based on the computed attention weights

$$\mathbf{z}_t = \sum_{r\in e_t} \alpha_i^{(t)} \cdot \mathbf{x}_i \tag{24}$$

where $\mathbf{z}_t$ denotes the feature representation of the tth hyperedge. Since each node may participate in multiple hyperedges, the updated representation of a node is obtained by performing a mean aggregation over the features of all hyperedges to which it belongs. This completes one round of structure-aware feature updating based on the spatial configuration of the hypergraph

$$\mathbf{x}_i^{(l+1)} = \frac{1}{|e_r|} \sum_{t\in e_r} \mathbf{z}_t \tag{25}$$

where e_r denotes the set of hyperedges incident to node v_i, and $|e_r|$ is the number of hyperedges containing v_i. Equations 23 and 24 form a closed loop of "intrahyperedge attention aggregation → hyperedge representation → node update," enabling both differentiated aggregation within each hyperedge and information propagation across hyperedges. From this perspective, the intrahyperedge attention mechanism can be interpreted as a learnable incidence strength.

Let $H \in \{0,1\}^{M\times E}$ denote the binary incidence matrix of a standard HGCN, where M and E are the numbers of nodes and hyperedges, respectively. We construct a weighted incidence matrix $\hat{H} \in \mathbb{R}^{M\times E}$, defined as

$$\hat{H}_{i,t} = \begin{cases} \alpha_i^{(t)}, & \text{if } v_i \in e_t \\ 0, & \text{otherwise} \end{cases} \tag{26}$$

where $\alpha_i^{(t)}$ is the attention coefficient of node v_i within hyperedge e_t. Furthermore, let $W = \mathrm{diag}(w_1, \ldots, w_E)$ denote the hyperedge weight matrix learned by the proposed ASSHW module.

Based on $\hat{H}$ and W, the vertex degree matrix D_v and the hyperedge degree matrix D_e are defined as

$$D_v = \mathrm{diag}\left(\hat{H} W \mathbf{1}_E\right) \tag{27}$$

$$D_e = \mathrm{diag}\left(\hat{H}^\top \mathbf{1}_M\right) \tag{28}$$

where $\mathbf{1}_E \in \mathbb{R}^{E\times 1}$ and $\mathbf{1}_M \in \mathbb{R}^{M\times 1}$ are all-ones vectors. Accordingly, the HGCEA layer can be formulated as an attention-parameterized hypergraph convolution

$$X^{(l+1)} = \sigma\left(D_v^{-\frac{1}{2}} \hat{H}\, W\, D_e^{-1}\, \hat{H}^\top\, D_v^{-\frac{1}{2}}\, X^{(l)}\, \Theta^{(l)}\right) \tag{29}$$

where $X^{(l)} \in \mathbb{R}^{M\times C_l}$, $\Theta^{(l)} \in \mathbb{R}^{C_l\times C_{l+1}}$, and $\sigma(\cdot)$ denotes a nonlinear activation function.

Compared with conventional hypergraph convolution or global attention mechanisms, the proposed module enables adaptive learning of node importance within each individual hyperedge, thereby dynamically adjusting the contribution of each node to the hyperedge-level feature representation. This facilitates fine-grained modeling of intrahyperedge structural relationships and further enhances the discriminative power and contextual awareness of node feature updates.

C. Key-Pixel Guided Transformer

When applying the ViT to HSIC, the self-attention mechanism performs pairwise interactions among all tokens, resulting in a quadratic computational complexity with respect to sequence length. To address this challenge, a KPFormer module is proposed. The goal is to reduce computational complexity while enhancing the model's ability to capture long-range spatial dependencies and focus on semantically critical regions.

As illustrated in Fig. 1, the KPFormer branch operates on the pixel-wise feature representations generated by the HGCEA branch, rather than directly on the raw HSI. First, the HGCEA-refined features are processed by a spectral convolution encoder module, SpectralConv1D, to obtain refined features $\mathbf{x}_i'$. This module integrates convolutional enhancement with residual connections, enabling spectral feature strengthening while preserving the structural cues encoded by hypergraph convolution. Unlike conventional ViTs that adopt a uniform grid-based patch extraction strategy, the proposed KPFormer performs key pixel selection on the pixel-wise features refined by the HGCEA branch. As shown on the left side of Fig. 4, the

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Fig. 4. Illustration of key-pixel selection and token refinement. The left part shows the selection of key pixels, where orange boxes denote selected pixels and white boxes represent unselected ones. The right part depicts the token construction and refinement process. By computing cosine similarity, highly relevant tokens are retained while irrelevant ones are filtered out, resulting in compact and discriminative token representations.

selection is guided by a fused importance score that combines structural attention and feature norm, aiming to improve the efficiency and discriminative capability of the transformer for HSIC. Specifically, the saliency score S_i for the ith pixel is defined as

$$S_i = \beta \cdot \|\mathbf{x}_i'\|_2 + (1-\beta)\sum_m Q_{im} a_m, \qquad \beta \in [0,1] \tag{30}$$

where $\|\mathbf{x}_i'\|_2$ denotes the feature norm of the ith pixel after spectral convolutional enhancement applied to the HGCEA output, and $\sum_m Q_{im} a_m$ is a weighted summation of the hyperpixel attention attributed to pixel i, indicating the structural contextual importance of the pixel. In this formulation, a_m denotes the attention score of the mth superpixel, and $\beta \in [0,1]$ is a fusion coefficient that balances spectral intensity and structural context contributions. Based on the computed saliency scores S_i, the p pixels with the highest scores are selected to form the key pixel set $\mathcal{S}_{\text{key}}$. Conventional key-pixel selection methods typically rely on backpropagation and require strong supervision, which may introduce bias toward dominant classes or lead to overfitting in class-imbalanced scenarios. In contrast, the proposed fused scoring scheme offers two distinct advantages: 1) it alleviates potential bias and overfitting by avoiding dependence on class-specific confidence or gradient signals and 2) it provides more interpretable guidance by jointly leveraging local structure-aware attention and spectral discriminability. As a result, the proposed strategy demonstrates strong robustness and efficiency, particularly in low-label or weakly supervised settings.

To overcome the limitations of fixed grid-based patch tokens in texture nonuniform regions and to better capture highly heterogeneous features such as object boundaries and local variations, KPFormer constructs content-adaptive token sequences centered on each selected key pixel. For each selected key pixel $v_p \in \mathcal{S}_{\text{key}}$, KPFormer constructs a dynamic token sequence by identifying the top-q most similar pixels within its spatial neighborhood, as shown on the right side of Fig. 4. The similarity is measured based on feature affinity, such as cosine similarity. This localized grouping allows the model to form content-adaptive token representations that capture fine-grained spectral–spatial correlations centered around each key pixel

$$T_p = \left\{ \mathbf{x}_q' \;\middle|\; v_q \in \mathcal{N}_p \cup \{v_p\} \right\} \tag{31}$$

where x_q' denotes the feature vector of the qth pixel within the selected neighborhood, and $\mathcal{N}_p$ represents the index set of the $q-1$ most similar pixels to the key pixel v_p.

After token construction, a CAF module is introduced to replace the standard feedforward network and enhance global awareness across tokens. Unlike conventional FFN modules, CAF introduces global modulation via semantic gating, which dynamically scales each token's channel response. This enhances intertoken coherence and improves robustness to spatial heterogeneity. Specifically, CAF first performs global average pooling along the channel dimension to extract a compact semantic vector, which is then used to modulate the response strength of each token. This mechanism improves the model's ability to distinguish target regions and increases robustness to local variations. Formally, each token group T_p is first linearly projected and embedded with positional information, and then fed into a transformer encoder to model contextual dependencies among tokens. The feature embedding of node p can be expressed as

$$\mathbf{z}_p = T_p + W_2 \cdot \text{ReLU}\left(W_1\left(T_p \odot g\right) + B_1\right) + B_2 \tag{32}$$

where $g = \text{AvgPool1D}(T)$ denotes the global contextual vector obtained by performing 1-D average pooling across all tokens along the channel dimension. The symbol $\odot$ represents element-wise (channel-wise) multiplication, which is used to perform channel-wise modulation of token features. W and B correspond to the weight matrix and bias vector of the linear transformation layer, respectively.

Overall, the proposed KPFormer introduces a structurally guided and content-adaptive token construction strategy built upon HGCEA-refined features. This design effectively alleviates the quadratic complexity of ViT while enhancing

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class-specific discrimination in heterogeneous hyperspectral scenes.

D. Residual Complement Fusion Network

To ensure semantic consistency across both key and nonkey pixels, and mitigate the sparsity introduced by selective token inference, we propose the RCFN. The HGCEA branch provides superpixel-level structural features from the HGCN, while the KPFormer branch contributes high-level semantic features for key pixels via the ViT. The proposed RCFN module integrates the strong semantic modeling capacity of the transformer with the global structural awareness of the HGCN, enabling complementary feature propagation from key pixels to the entire image. This design facilitates semantic enhancement in nonkey regions and improves the overall classification performance and robustness, particularly for edge pixels. Ultimately, it supports unified prediction across all image pixels.

Specifically, the RCFN module adopts a differentiated strategy for distinct pixel types by separately processing key pixels $\mathcal{S}_{\text{key}}$ and nonkey pixels $\mathcal{S}_{\text{main}}$, ensuring that the final prediction maintains both local detail and global consistency. For key pixels $\mathcal{S}_{\text{key}}$, the features from the ViT are fused with auxiliary information from the HGCN to capture enriched semantic context. For nonkey pixels $\mathcal{S}_{\text{main}}$, the upsampled output of the HGCN is directly utilized as the prediction source. The residual enhancement formulation is defined as follows:

$$\hat{\mathbf{X}}_i = \begin{cases} \mathbf{X}_{\text{KPFormer}} + \lambda \cdot \mathbf{X}_{\text{HGCEA}}, & i \in \mathcal{S}_{\text{key}} \\ (1+\lambda) \cdot \mathbf{X}_{\text{HGCEA}}, & i \in \mathcal{S}_{\text{main}} \end{cases} \tag{33}$$

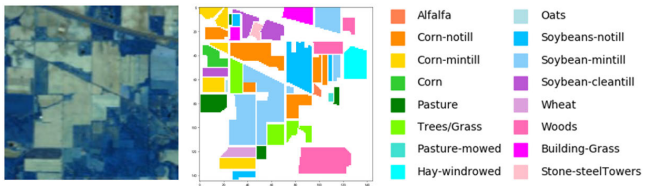
where $\hat{\mathbf{X}}i$ denotes the final fused feature representation of the ith pixel. $\mathbf{X}_{\text{KPFormer},i}$ represents the output feature from the KPFormer branch for the corresponding key pixel, while $\mathbf{X}_{\text{HGCEA},i}$ refers to the prediction from the HGCEA branch for the ith pixel. The coefficients λ are weighting coefficients used to balance the information fusion between the ViT-based KPFormer and the hypergraph-based HGCEA output. Subsequently, the fused feature matrix $\hat{\mathbf{X}}$ is fed into a multilayer perceptron (MLP) classifier to produce the category probability distribution for each pixel

$$\mathbf{Y}_i = \text{Softmax}\left(W \cdot \hat{\mathbf{X}}_i + B\right). \tag{34}$$

Finally, to train the network, a masked multiclass cross-entropy loss is employed to supervise the pixel-wise classification. Considering that HSIC is typically conducted under sparse supervision, only a small subset of pixels is annotated for training. To avoid introducing noisy gradients from unlabeled regions, a binary mask is applied to ensure that only labeled pixels participate in loss computation. The loss function is defined as follows:

$$\mathcal{L}_{\text{mask}} = -\sum_{i=1}^{N}\sum_{c=1}^{C} M_{i,c} \cdot \mathbf{Y}'_{i,c} \cdot \log\left(\mathbf{Y}_{i,c}\right) \tag{35}$$

where $\mathbf{Y}'_{i,c}$ denotes the ground-truth label of the ith pixel for the cth class, while $\mathbf{Y}_{i,c}$ represents the predicted class probability output by the model. $M_{i,c}$ is a binary mask coefficient, where $M_{i,c} = 1$ indicates that the pixel is labeled and contributes to the loss, and $M_{i,c} = 0$ otherwise. This formulation can be viewed as a masked extension of the standard cross-entropy loss, which is more suitable for HSIC with partial pixel-level annotations.



(a) (b)

Fig. 5. Plot of IP dataset. (a) Pseudo-color plot and (b) ground truth plot, sample label colors on the right side.

The proposed structure fuses the globally contextual representations from the transformer with the structurally enriched features from the HGCN for key pixels, resulting in more discriminative outputs. The fused feature map $\hat{\mathbf{X}} \in \mathbb{R}^{H\times W\times C}$ can be directly reshaped into a full-resolution pixel-level classification map. This output is subsequently passed through a final MLP classifier for refinement, further improving boundary smoothness and interclass separability.

IV. Experimental Results

In this section, we evaluate the effectiveness of the proposed methodology through comprehensive experiments conducted on four widely used hyperspectral datasets: Indian Pines (IP), Pavia University (PU), Salinas (SA), and WHU-Hi-HongHu (HongHu). All classification models were implemented using the PyTorch deep learning framework. The experiments were performed on a workstation equipped with an Intel[1] Xeon[1] Platinum 8151 CPU @ 3.40 GHz and an NVIDIA Quadro GV100 GPU.

A. Description of Experimental Dataset

The IP dataset was collected using the Airborne Visible/Infrared Imaging Spectrometer over a test site in Indiana, USA. This HSI has a spatial resolution of 145 × 145 pixels and covers an area of approximately 5.76 km^2. Initially comprising 224 spectral bands within the 0.4–2.5-μm range, the dataset is typically reduced to 200 effective bands by excluding those severely impacted by water vapor absorption. The scene is dominated by agricultural and semi-natural land types, including crops, woodlands, transportation infrastructure, and man-made structures, as shown in Fig. 5. A total of 16 land-cover classes are defined, and the number of selected labeled samples per class is summarized in Table I.

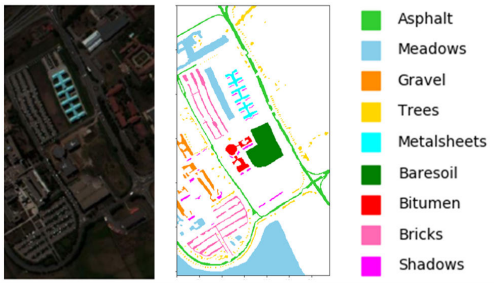
The PU dataset was acquired by the ROSIS-3 hyperspectral imaging sensor during an airborne campaign conducted over the city of Pavia, Italy. The image covers a spatial extent of 610 × 340 pixels with a ground sampling distance of 1.3 m. The original data consists of 115 spectral bands spanning the wavelength range from 430 to 860 nm. In practice, 12 noisy bands—primarily located in water absorption regions—are

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TABLE I
CATEGORY-WISE SAMPLE DISTRIBUTION IN THE IP DATASET

No.	Class	Train	Val.	Test
1	Alfalfa	15	1	30
2	Corn-notill	30	1	1397
3	Corn-mintill	30	3	797
4	Corn	30	2	205
5	Pasture	30	3	450
6	Trees/Grass	30	3	697
7	Pasture-mowed	15	1	12
8	Hay-windrowed	30	1	447
9	Oats	15	1	4
10	Soybeans-notill	30	1	941
11	Soybeans-mintill	30	3	2422
12	Soybeans-cleantill	30	2	561
13	Wheat	30	2	173
14	Woods	30	3	1232
15	Residential-Grass-Mixed	30	1	355
16	Stone–Steel Towers	30	2	61
Total		**435**	**30**	**9514**



(a) (b)

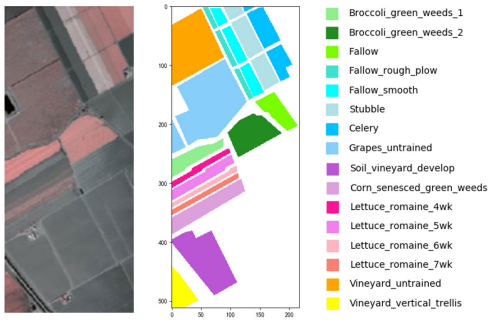
Fig. 6. Plot of PU dataset. (a) Pseudo-color plot and (b) ground truth plot, sample label colors on the right side.

TABLE II
CATEGORY-WISE SAMPLE DISTRIBUTION IN THE PU DATASET

No.	Class	Train	Val.	Test
1	Asphalt	30	4	6597
2	Meadows	30	12	18607
3	Gravel	30	3	2066
4	Trees/Grass	30	1	3033
5	Metal Sheets	30	4	1311
6	Bare Soil	30	2	4997
7	Bitumen	30	2	1298
8	Bricks	30	1	3651
9	Shadows	30	1	916
Total		**270**	**30**	**42476**

typically removed, resulting in 103 effective spectral bands used for analysis. The scene contains nine land-cover categories, including asphalt, meadows, and bare soil, as shown in Fig. 6. The number of selected samples for each class used in this study is summarized in Table II.

The SA dataset was provided by the Hyperspectral Image Analysis Group in collaboration with the NSF-supported National Center for Airborne Laser Mapping at the University of Houston. It comprises 144 spectral bands spanning wavelengths from 380 to 1050 nm. The image covers an area of 349 × 1905 pixels, with a spatial resolution of 2.5 m. This urban scene includes a wide variety of land-cover types, such as healthy grass, stressed grass, artificial turf, and airport runways, as shown in Fig. 7. The class-wise distribution of selected labeled samples is detailed in Table III.



(a) (b)

Fig. 7. Plot of the SA dataset. (a) Pseudo-color plot and (b) ground truth plot, sample label colors on the right side.

TABLE III
CATEGORY-WISE SAMPLE DISTRIBUTION IN THE SA DATASET

No.	Class	Train	Val.	Test
1	Broccoli_green_weeds_1	30	2	1977
2	Broccoli_green_weeds_2	30	2	3694
3	Fallow	30	1	1945
4	Fallow_rough_plow	30	2	1362
5	Fallow_smooth	30	1	2647
6	Stubble	30	3	3926
7	Celery	30	2	3547
8	Grapes_untrained	30	3	11238
9	Soil_vineyard_develop	30	3	6170
10	Corn_senesced_green_weeds	30	1	3247
11	Lettuce_romaine_4wk	30	1	1037
12	Lettuce_romaine_5wk	30	2	1895
13	Lettuce_romaine_6wk	30	1	885
14	Lettuce_romaine_7wk	30	2	1038
15	Vineyard_untrained	30	3	7235
16	Vineyard_vertical_trellis	30	1	1776
Total		**480**	**30**	**53619**



(a) (b)

Fig. 8. Plot of HongHu dataset. (a) Pseudo-color plot and (b) ground truth plot, sample label colors on the right side.

The HongHu dataset was collected using a Headwall Nano-Hyperspec imaging sensor during an airborne campaign conducted over Honghu City, Hubei, China. The image comprises 940 × 475 pixels with a spatial resolution of approximately 0.043 m. It includes 270 spectral bands spanning wavelengths from 400 to 1000 nm. The scene represents a complex agricultural area characterized by diverse crop types, including Chinese cabbage, cabbage, Brassica chinensis, and small Brassica chinensis, as shown in Fig. 8. The distribution of labeled samples for each category is summarized in Table IV.

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TABLE IV
CATEGORY-WISE SAMPLE DISTRIBUTION IN THE HONGHU DATASET

No.	Class	Train	Val.	Test
1	Red roof	30	1	1977
2	Road	30	2	3694
3	Bare soil	30	1	1945
4	Cotton	30	2	1362
5	Cotton firewood	30	1	2647
6	Rape	30	1	3926
7	Chinese cabbage	30	1	3547
8	Pakchoi	30	3	11238
9	Cabbage	30	2	6170
10	Tuber mustard	30	2	3247
11	Brassica parachinensis	30	1	1037
12	Brassica chinensis	30	2	1895
13	Small Brassica chinensis	30	1	885
14	Lactuca sativa	30	1	1038
15	Celtuce	30	2	7235
16	Film-covered lettuce	30	1	1776
17	Romaine	30	1	1776
18	Carrot	30	1	1776
19	White radish	30	1	1776
20	Garlic sprout	30	1	1776
21	Broad bean	30	1	1776
22	Tree	30	1	1776
Total		**660**	**30**	**386003**

B. Experiment Settings

1) Evaluation Metrics: To evaluate the classification performance, three widely adopted quantitative metrics were employed: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (κ). Specifically, OA is defined as the ratio of correctly predicted samples to the total number of test samples, thereby indicating the global classification effectiveness. AA is calculated by averaging the individual accuracies of all categories, which ensures a more balanced evaluation across classes. The κ, which accounts for the possibility of random agreement, offers a more reliable measurement of the consistency between the predicted labels and the ground truth. In addition to these quantitative indicators, a qualitative assessment was conducted by visualizing the classification maps, facilitating an intuitive evaluation of spatial coherence and boundary discrimination achieved by the proposed method.

2) Comparison Methods and Sample Selection: To comprehensively assess the performance of the proposed HGTFN model, ten representative classification methods were selected, encompassing various technical paradigms and development stages. The comparison includes a classical machine learning algorithm—SVM [47]; two traditional deep learning models—3-D-CNN [48] and SSRN [17]; two advanced transformer-based approaches—multilevel graph hybrid transformer (MGHT) [49], and SF [24]; as well as five state-of-the-art graph-based methods—HGCN (HGCN-MHF) [41], convolution-enhanced graph convolutional network (CEGCN) [50], attention-based multihop graph and multiscale convolutional fusion network (AMGCFN) [51], CNN–GAT-based weighted feature fusion network (WFCG) [52], and graph convolution–transformer fusion network (GTFN) [53]. This comprehensive selection of baselines ensures a robust and multifaceted comparison, enabling a thorough evaluation of the proposed HGTFN framework across diverse technical dimensions.

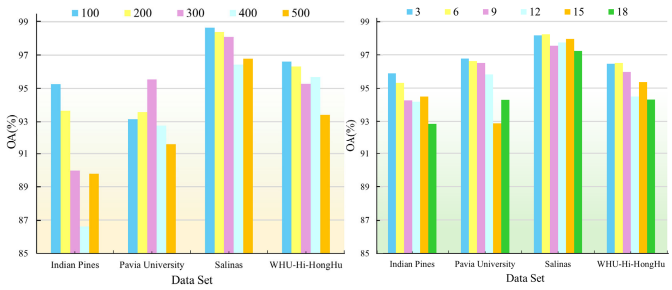


Fig. 9. Left panel shows the OA for different segmentation scales in the four benchmark datasets. The right panel shows the OA for different numbers of KNN hyperedge neighbors in the four benchmark datasets.

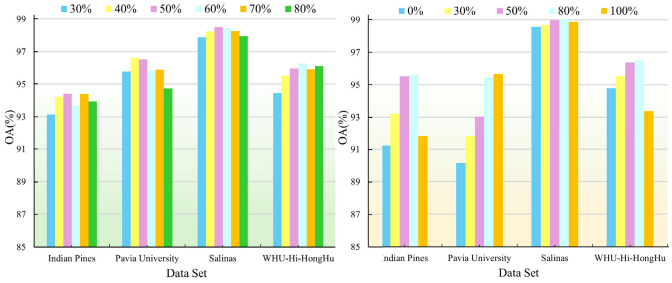


Fig. 10. Left panel shows the OA for different key pixel selection ratios in the four benchmark datasets. The right panel shows the OA for different residual fusion ratios in the four benchmark datasets.

3) Parameter Settings and Implementation Details: To ensure a fair comparison, a consistent experimental configuration was applied across all methods. Each model was trained for 300 epochs with a fixed learning rate of 0.001. For training, 30 samples were randomly selected per class. If a class contained a limited number of labeled samples, the number of training samples was constrained to be less than half of the total available labeled samples. For validation, at least one sample was randomly selected from each class, resulting in a total of 30 validation samples. All remaining samples were used for testing. The reported results represent the average performance over ten independent runs, with different training and validation sets randomly selected in each run. In both the comparative experiments and the HGTFN framework, the number of attention heads was set to 4, the patch size was set to 5×5, the patch batch size was 9, and the hyperparameters α and β were set to 0.6 and 0.5, respectively.

C. Hyperparameter Analysis

In this section, we analyze the impact of several key hyperparameters on the model's performance. Specifically, we examine: the SLIC segmentation scale parameter (λ) and the number of neighbors in KNN-based hyperedges (n_{knn}) in the SMHGC module; the key pixel selection ratio (n_p, where $n_p = p/N$) in the KPFormer module; the residual fusion ratio (λ) in the RCFN module; and the network depth (l) of each submodule.

1) Effect of Segmentation Scale λ: To evaluate the effect of superpixel granularity on classification performance, we conducted experiments under different SLIC segmentation

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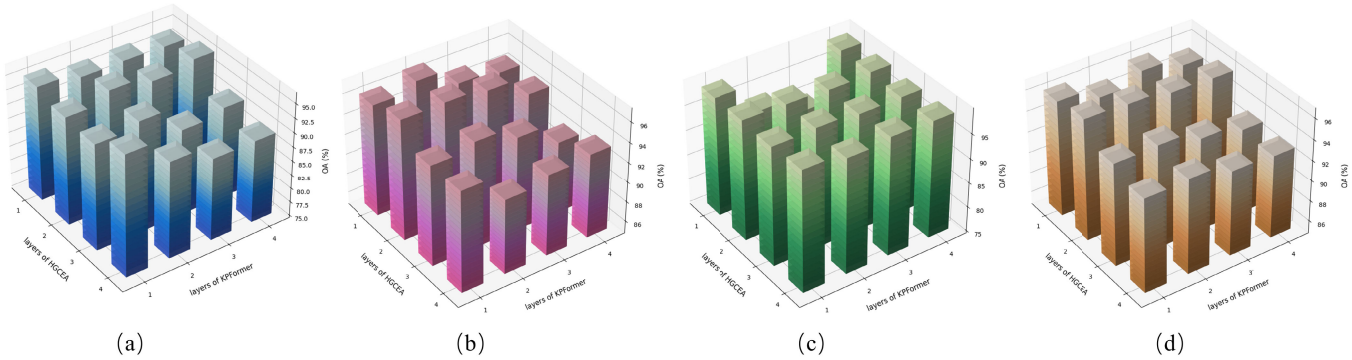


Fig. 11. OA under different combinations of HGCEA and KPFormer layers across four datasets. (a) IP. (b) PU. (c) SA. (d) HongHu. The horizontal axes denote the number of layers in HGCEA and KPFormer modules, respectively.

TABLE V
EXPERIMENTAL RESULTS COMPARING DIFFERENT ALGORITHMS ON THE IP DATASET

Class	SVM	3D-CNN	SSRN	SF	MGHT	CEGCN	HGCN-MHF	AMGCFN	WFCG	GTFN	HGTFN
1	39.04±1.88	77.83±7.89	92.12±6.50	90.65±3.23	90.63±2.55	93.75±0.54	**99.86±0.12**	85.71±1.44	93.16±1.95	96.77±2.72	95.65±2.17
2	61.02±6.51	83.61±4.75	86.63±8.00	67.41±7.92	81.34±0.63	84.52±4.62	84.87±3.68	84.07±2.58	87.73±2.32	89.20±1.55	**92.13±0.31**
3	71.68±3.15	87.59±2.02	75.55±9.25	80.75±3.56	92.00±1.71	83.03±5.98	84.75±4.96	**99.52±0.38**	96.25±0.31	81.50±3.46	83.73±0.98
4	30.15±7.68	79.67±6.41	81.02±4.11	**96.59±2.59**	92.45±1.47	84.66±1.24	**99.72±0.20**	79.20±1.35	86.67±0.16	**99.52±0.22**	95.23±0.95
5	84.16±4.70	79.31±4.81	89.97±4.55	95.67±1.48	92.70±1.80	91.57±2.05	94.22±2.42	94.58±2.83	85.84±4.57	94.26±1.08	**96.67±1.92**
6	90.09±1.90	89.08±5.01	92.90±3.21	92.58±3.95	91.91±2.34	**99.71±0.13**	97.42±0.74	97.03±1.45	92.46±0.15	96.14±3.17	97.96±0.75
7	52.65±3.92	76.04±7.64	94.30±4.21	90.86±2.17	88.34±3.29	99.23±0.22	99.42±0.21	81.23±1.75	92.38±4.31	92.31±1.28	**100.00±0.00**
8	89.30±2.37	79.92±7.62	95.07±3.76	95.74±1.24	94.00±0.68	99.55±0.34	98.76±0.12	99.12±0.09	83.24±0.12	98.88±1.21	**99.74±0.06**
9	39.42±8.32	68.65±1.60	99.25±0.34	92.99±1.01	94.44±0.47	98.79±0.41	99.07±0.54	**100.00±0.00**	95.24±3.69	**100.00±0.00**	**100.00±0.00**
10	62.10±4.84	85.65±5.79	83.69±7.76	63.03±7.22	85.92±4.16	84.23±1.62	91.71±1.58	87.68±2.94	**92.73±0.33**	93.63±2.06	91.15±1.62
11	79.94±4.78	87.16±5.01	90.53±4.22	77.17±5.82	91.62±1.43	96.82±0.66	96.32±1.82	97.62±2.06	98.47±0.31	81.65±2.24	**99.40±0.82**
12	59.97±1.42	81.36±4.04	89.97±4.71	91.22±2.82	87.70±3.47	89.34±2.95	92.51±6.07	77.28±3.74	86.11±0.18	84.90±1.96	**94.03±1.85**
13	90.05±3.92	78.91±6.28	98.62±1.25	93.12±1.43	93.22±0.68	98.46±0.75	99.42±0.35	93.26±0.93	**99.68±0.09**	98.86±0.33	98.24±0.93
14	89.81±3.25	79.73±7.96	99.16±0.12	92.82±1.16	93.33±0.74	98.91±0.97	**100.00±0.00**	99.67±0.18	84.87±0.13	99.03±0.48	99.82±0.51
15	48.73±1.02	72.79±6.97	88.64±5.14	77.02±6.12	92.78±5.09	99.43±0.37	**100.00±0.00**	99.73±0.21	92.59±0.69	97.47±1.76	**100.00±0.00**
16	80.13±1.65	80.01±4.26	98.86±1.07	95.94±1.04	90.55±4.42	98.38±0.55	**100.00±0.00**	97.80±2.09	89.57±5.68	**100.00±0.00**	**100.00±0.00**
OA (%)	73.97±2.83	76.25±3.13	87.53±3.32	81.54±2.56	91.32±1.86	92.51±2.35	93.99±1.93	93.40±3.94	91.74±2.71	90.09±3.26	**95.73±1.14**
AA (%)	66.82±1.69	76.71±5.52	89.12±4.98	87.10±3.50	91.43±1.99	94.49±1.41	96.31±1.54	92.02±1.19	90.49±1.97	94.01±2.28	**96.48±1.05**
Kappa	70.29±2.98	76.44±4.88	85.76±3.36	79.02±3.15	90.10±2.13	91.36±2.25	93.12±2.46	92.47±3.26	90.58±2.79	88.73±3.30	**95.06±1.16**

scales $\lambda \in \{100, 200, 300, 400, 500\}$ across four benchmark datasets. The corresponding OA is shown on the left side of Fig. 9. The SLIC scale parameter λ governs superpixel size and compactness. Smaller λ increases detail and boundary precision at the cost of noise/computation; larger λ reduces complexity but risks oversmoothing and blurred boundaries. As observed, the segmentation scale significantly influences the model's performance. On the IP dataset, $\lambda = 100$ achieves the highest OA, while larger scales (e.g., $\lambda = 400, 500$) result in noticeable degradation, likely due to over-smoothing and loss of fine structures. For PU, the best performance is observed at $\lambda = 300$, indicating that a moderate segmentation scale better preserves spatial structures and semantic boundaries. For SA and HongHu, the OA remains relatively stable for $\lambda = 100$ to 200, but tends to decrease as λ increases further. Therefore, for these four datasets, λ is set to 100, 300, 100, and 100, respectively, to obtain the highest OA.

2) Effect of the Neighborhood of KNN n_{knn}: To evaluate the effect of KNN-based hyperedge connectivity, we vary the number of neighbors n_{knn} from 3 to 18 on four benchmark datasets. The result is shown in the right panel of Fig. 9. The parameter n_{knn} determines the density of the hypergraph. A small value may limit contextual propagation due to sparse connections, while a large value may introduce noise and reduce discriminability. Experimental results show that $n_{\text{knn}} = 3$ or 6 consistently yields better performance. In particular, accuracy drops when $n_{\text{knn}} > 12$ on IP, PU and HongHu. The SA dataset is less sensitive, likely due to its large homogeneous regions. To save computational overhead, we set n_{knn} to 6 to achieve a good balance between accuracy and stability.

3) Effect of Key Pixel Ratio n_p: The left panel of Fig. 10 illustrates the impact of different key pixel selection ratios (ranging from 30% to 80%) on classification performance across four benchmark datasets. The key pixel ratio determines the proportion of pixels passed into the transformer branch for global modeling. A low ratio may fail to capture sufficient global context, while an excessively high ratio increases computational overhead and may introduce redundant or noisy tokens, weakening the discriminative capacity. As observed, performance generally improves with increasing ratio up to a point. The OA rises from 30% to 50% across most datasets, and peaks around 50%–60%. Beyond this point, gains saturate or slightly degrade, especially on IP and PU. Therefore, a ratio of 50% is selected as the optimal setting, striking a balance between performance and efficiency. Therefore, for these four datasets, we chose a ratio of 50% as the optimal setting.

4) Effect of Residual Fusion Ratio λ: Residual formulation 33 offers a controllable and interpretable mechanism for propagating semantic information from transformer-enhanced key pixels to the rest of the image, which is especially beneficial under sparse supervision. The right panel of Fig. 10 presents the classification performance under different residual fusion

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TABLE VI

EXPERIMENTAL RESULTS COMPARING DIFFERENT ALGORITHMS ON THE PU DATASET

Class	SVM	3D-CNN	SSRN	SF	MGHT	CEGCN	HGCN-MHF	AMGCFN	WFCG	GTFN	HGTFN
1	83.82±1.56	84.01±0.65	91.08±2.49	66.22±6.89	97.40±0.14	**99.24±0.18**	89.35±4.36	88.19±4.57	91.61±3.62	84.21±0.73	90.13±1.92
2	94.75±0.24	83.32±2.75	**100.00±0.00**	78.06±5.43	97.93±0.06	97.50±2.61	97.98±1.26	98.92±0.80	**100.00±0.00**	95.05±0.29	98.28±1.57
3	56.59±4.22	81.75±1.56	80.46±3.26	93.13±2.35	82.09±0.17	72.67±5.51	**96.11±0.46**	70.79±5.89	92.16±1.28	78.44±0.70	94.36±1.56
4	74.34±0.96	84.62±2.68	86.33±2.69	53.75±8.12	81.62±0.28	88.44±2.65	78.68±8.56	**95.60±2.46**	72.08±6.44	89.72±0.83	86.06±5.81
5	72.62±0.65	81.45±4.96	94.87±1.23	98.70±0.46	**99.83±0.13**	96.16±0.19	92.27±1.46	99.55±0.28	99.69±0.09	98.94±0.46	94.58±1.29
6	71.06±1.63	75.46±9.68	82.45±0.45	97.43±1.23	78.81±0.19	**99.56±0.17**	99.41±0.12	98.94±0.97	97.96±1.35	99.12±0.23	99.27±0.46
7	68.03±2.58	76.41±1.14	95.76±1.20	85.69±1.49	94.97±0.10	93.76±4.72	97.98±1.28	92.79±1.22	97.02±1.24	97.62±1.89	**98.66±1.28**
8	81.87±1.96	84.21±0.88	90.06±0.96	86.11±3.57	94.84±0.39	91.09±3.19	89.64±3.47	**99.01±0.30**	88.29±6.24	78.75±0.79	96.49±1.60
9	68.99±3.20	82.20±0.42	81.29±3.69	36.75±12.34	99.71±0.22	94.91±1.25	75.02±6.84	69.17±7.25	**99.89±0.08**	96.95±1.36	87.51±6.05
OA (%)	83.73±0.94	82.29±1.97	93.05±1.79	78.17±5.34	93.40±0.81	95.40±2.09	94.16±1.96	94.83±2.98	94.97±2.19	91.50±0.37	**95.58±1.69**
AA (%)	74.67±1.16	81.49±2.02	89.14±1.27	77.32±4.12	91.91±0.86	92.59±2.12	90.72±2.34	90.35±1.35	93.19±1.57	90.98±0.76	**93.93±1.87**
Kappa	78.44±0.79	76.54±1.66	90.82±1.49	72.27±4.53	91.25±1.12	93.84±7.28	92.02±2.08	93.17±2.20	93.33±2.76	88.81±0.43	**94.13±1.76**

TABLE VII

EXPERIMENTAL RESULTS COMPARING DIFFERENT ALGORITHMS ON THE SA DATASET

Class	SVM	3D-CNN	SSRN	SF	MGHT	CEGCN	HGCN-MHF	AMGCFN	WFCG	GTFN	HGTFN
1	98.94±1.01	97.79±0.13	99.51±0.34	98.56±0.95	99.80±1.01	96.15±1.72	99.84±0.16	95.03±0.88	98.62±0.65	**100.00±0.00**	**100.00±0.00**
2	92.33±1.70	94.55±1.72	99.54±0.35	94.62±0.11	99.76±0.13	99.96±0.06	99.85±0.05	**100.00±0.00**	**100.00±0.00**	99.95±0.02	**100.00±0.00**
3	94.75±0.42	92.48±1.21	99.63±0.16	98.47±1.36	99.85±0.17	99.93±0.05	99.94±0.01	97.56±0.79	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
4	98.91±0.31	98.63±0.14	99.53±0.25	98.33±0.99	99.01±0.44	**100.00±0.00**	99.63±0.14	82.72±3.28	99.92±0.06	99.49±0.09	95.85±0.96
5	96.27±0.93	90.08±0.73	96.04±3.26	97.65±1.98	98.31±1.07	97.41±1.72	98.18±0.35	97.11±0.71	**99.01±0.54**	93.35±3.56	98.18±0.26
6	97.49±0.58	96.73±1.27	99.48±0.32	99.43±0.21	99.32±0.06	99.39±0.60	**99.79±0.27**	97.51±0.10	98.99±0.50	98.77±2.00	99.76±0.12
7	95.22±0.43	97.74±0.14	99.35±0.55	98.29±1.53	96.81±0.37	99.86±0.15	99.85±0.14	98.45±0.64	**99.97±0.02**	99.89±0.10	99.85±0.06
8	65.59±1.75	72.31±0.59	92.24±5.36	75.12±8.33	88.45±1.48	90.23±1.77	95.40±7.46	95.13±0.85	88.36±4.22	85.26±5.78	**99.07±0.39**
9	83.96±1.34	95.23±0.86	97.79±0.37	94.68±3.64	98.23±0.58	95.25±0.31	95.71±3.68	**100.00±0.00**	95.59±0.45	99.45±0.37	**100.00±0.00**
10	81.38±4.38	83.84±0.97	95.81±3.78	84.87±3.15	91.35±1.36	**99.90±0.04**	87.13±8.15	97.18±0.57	97.50±0.58	94.74±5.05	98.77±0.53
11	94.08±1.59	97.24±0.28	99.59±0.24	93.68±0.31	98.19±1.48	99.43±0.30	99.51±0.32	93.79±0.36	98.78±1.54	**100.00±0.00**	**100.00±0.00**
12	98.78±0.79	96.79±0.04	99.19±0.02	99.83±1.68	99.87±0.12	99.88±0.03	**99.89±0.11**	81.88±2.45	98.86±0.24	99.42±0.05	99.83±0.10
13	96.43±0.39	97.22±0.14	98.09±1.33	97.45±1.52	94.00±2.94	98.83±0.39	99.77±0.13	**100.00±0.00**	99.81±0.05	99.86±0.16	95.27±0.89
14	98.00±0.54	95.53±1.55	99.06±0.38	95.81±0.68	**99.31±0.67**	97.83±1.10	99.51±0.32	96.05±0.37	98.72±1.24	98.65±0.99	96.32±0.32
15	75.58±0.45	83.19±3.13	79.81±1.77	97.78±2.98	92.86±3.13	88.79±3.41	74.93±3.56	86.52±1.46	90.12±3.25	87.07±3.35	**98.75±0.36**
16	95.99±0.37	92.01±1.15	98.45±1.02	99.59±0.34	99.28±0.58	99.61±0.11	98.53±1.08	98.88±0.39	95.94±0.78	98.71±1.21	**100.00±0.00**
OA (%)	85.18±0.45	88.26±1.39	94.67±4.34	91.13±1.89	95.33±2.11	95.79±0.53	94.71±1.68	95.17±0.06	95.13±1.24	94.32±0.59	**99.19±0.33**
AA (%)	91.81±1.07	92.65±0.86	97.06±2.29	95.26±1.58	97.15±1.05	97.77±0.37	97.09±1.03	94.77±0.62	97.39±0.77	97.21±0.29	**98.85±0.57**
Kappa	83.49±0.65	86.93±0.86	94.07±2.47	90.04±1.67	94.80±1.55	95.31±0.59	94.10±1.58	94.62±0.07	94.60±1.56	93.88±0.68	**99.10±0.43**

TABLE VIII

EXPERIMENTAL RESULTS COMPARING DIFFERENT ALGORITHMS ON THE HONGHU DATASET

Class	SVM	3D-CNN	SSRN	SF	MGHT	CEGCN	HGCN-MHF	AMGCFN	WFCG	GTFN	HGTFN
1	94.09±4.99	92.44±2.64	93.12±2.98	93.51±1.58	92.43±2.52	93.49±11.69	95.76±1.07	93.19±1.86	89.77±7.98	93.18±1.41	**97.86±1.36**
2	87.04±7.14	91.55±1.86	85.04±1.58	90.30±1.46	**95.73±1.44**	87.29±9.80	91.95±6.44	94.42±1.44	92.49±6.53	91.85±2.48	82.81±1.06
3	93.99±6.13	83.80±2.54	96.47±1.43	87.44±2.65	83.33±1.42	**99.41±0.53**	96.10±3.17	81.04±5.40	89.96±4.82	82.21±6.45	96.81±0.54
4	74.10±5.97	97.71±1.67	91.28±1.19	86.71±2.74	98.25±0.39	95.95±2.90	98.22±1.06	**99.60±0.09**	95.14±2.30	99.41±0.32	98.42±0.31
5	80.11±5.56	88.19±1.95	86.18±2.43	97.77±1.08	92.07±2.54	98.10±1.20	97.88±1.98	93.70±9.16	96.56±2.62	96.45±1.96	**100.00±0.00**
6	76.06±2.55	78.02±5.51	91.63±1.80	96.01±1.55	91.08±2.15	95.98±3.48	97.59±1.18	91.99±4.17	**98.06±1.52**	92.84±1.34	96.28±1.23
7	91.71±5.31	**98.80±0.73**	83.59±7.64	61.26±6.28	89.83±9.83	78.21±8.83	63.78±6.49	97.43±1.28	91.66±5.75	92.95±4.36	94.52±2.34
8	72.41±6.14	74.96±6.79	63.38±9.82	88.45±2.29	83.22±1.54	86.16±14.91	95.70±1.95	78.19±11.37	84.21±9.79	75.73±6.55	**99.14±1.28**
9	**98.94±0.02**	91.78±1.40	92.53±2.49	98.85±1.14	93.64±1.62	91.85±4.59	95.33±2.74	97.37±1.02	92.53±2.63	82.83±1.31	99.18±1.14
10	62.29±7.60	70.79±3.71	86.16±4.93	97.89±0.95	95.41±1.90	87.27±1.51	86.49±8.59	90.84±3.98	86.16±4.93	**95.59±1.49**	92.75±1.35
11	74.61±7.50	70.76±11.32	72.67±9.84	82.27±3.19	**93.40±9.65**	56.43±9.82	87.85±3.59	79.01±7.90	85.67±4.52	67.70±8.74	91.77±1.27
12	76.81±6.34	83.89±9.05	80.70±10.68	66.14±6.71	90.12±7.17	89.97±8.77	80.70±3.72	82.54±7.30	86.70±8.64	79.44±9.64	**97.52±0.56**
13	70.93±5.06	67.63±8.65	86.48±8.10	71.57±8.49	86.89±0.35	86.11±9.87	72.29±6.76	87.89±0.91	79.48±6.86	69.57±9.56	**90.49±2.48**
14	87.93±2.16	66.82±6.12	89.10±5.94	91.99±3.59	83.00±1.61	87.76±9.93	87.86±1.38	91.74±1.49	82.10±7.61	83.36±5.21	**94.27±0.76**
15	77.52±6.09	98.44±0.13	83.77±4.30	96.45±1.34	97.08±1.66	98.45±0.64	99.89±0.09	99.00±0.31	**99.77±0.16**	99.56±0.42	98.02±0.36
16	92.31±2.06	93.89±1.76	93.19±2.86	94.08±2.65	**95.64±2.86**	93.27±5.49	97.42±1.72	88.24±0.79	93.19±4.53	87.65±5.72	90.23±0.62
17	82.90±5.53	83.08±5.45	97.02±0.52	89.71±3.46	92.59±2.59	94.49±1.27	99.59±0.31	92.19±4.25	91.02±8.28	92.13±0.42	**97.67±1.25**
18	96.25±1.35	89.79±5.03	82.21±1.53	97.08±0.41	96.52±1.88	98.35±1.01	96.83±3.08	94.19±2.65	92.21±4.30	**99.11±0.21**	96.16±0.56
19	74.55±6.04	82.20±6.73	87.45±5.36	96.15±1.95	87.04±2.83	93.15±1.74	89.16±4.84	91.23±1.60	94.04±2.83	**97.04±0.58**	95.56±2.73
20	79.58±5.29	95.62±1.28	94.68±1.48	99.94±0.05	97.08±3.66	94.62±3.36	96.73±3.24	98.75±1.08	94.68±11.85	99.03±0.53	**99.55±0.04**
21	80.52±8.62	92.01±1.19	84.08±5.55	**99.92±0.03**	97.73±2.43	86.72±2.63	99.30±0.26	93.47±4.13	91.19±3.67	95.07±2.38	99.43±0.32
22	94.12±1.09	93.57±2.03	92.75±2.75	**99.89±0.49**	93.46±1.49	98.05±0.10	99.52±0.15	96.77±2.81	94.75±3.80	89.94±0.05	96.86±1.59
OA (%)	78.83±3.40	87.69±4.56	89.47±2.18	89.13±3.46	94.82±0.59	92.30±1.37	92.63±1.78	94.22±1.54	92.54±1.34	92.21±1.26	**96.56±0.86**
AA (%)	82.67±3.09	83.44±4.13	86.98±2.02	91.94±2.58	93.62±1.97	90.50±1.14	92.09±1.94	91.49±1.27	90.97±1.73	89.07±0.57	**95.70±1.41**
Kappa	73.36±4.69	84.46±5.67	86.37±2.73	88.22±3.69	93.52±2.37	90.39±2.60	90.71±1.55	92.73±2.65	90.67±2.73	90.23±4.82	**95.66±1.19**

ratios (ranging from 0% to 100%) across the four datasets. This ratio controls the strength of residual compensation for nonkey pixels. When set to 0%, no residual information is fused, which may lead to missing local semantics. Conversely, a ratio of 100% overly relies on residual features, possibly weakening the discriminative effect of transformer-enhanced key pixels. The results show that moderate ratios (around 80%) yield the best performance on most datasets. Notably, performance declines at 100% for IP and HongHu, confirming the risk of over-compensation. Thus, a residual fusion ratio of 80% is adopted as the default setting.

5) Effect of Subnetwork Depths l: To assess the influence of model depth, we systematically varied the number of layers in both the HGCEA and KPFormer modules from 1 to 4 and evaluated their combinations across four benchmark datasets. As shown in Fig. 11, OA is consistently enhanced by moderate increases in both modules, confirming the complementary strengths of hypergraph-based local representation and transformer-based global modeling. Notably, performance peaks are generally achieved when both modules are set to two or three layers. Excessive stacking beyond four layers tends to yield diminishing returns or slight performance drops, likely

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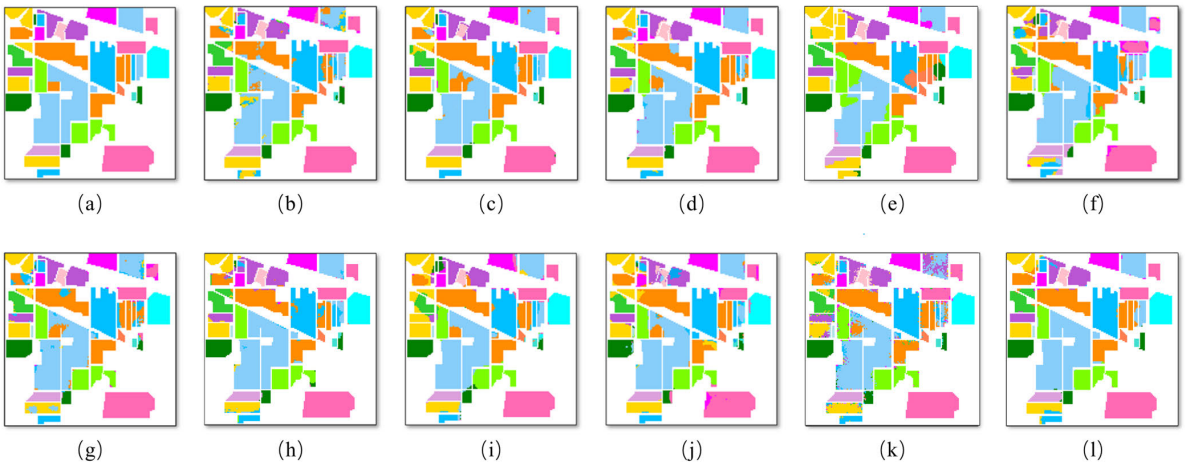


Fig. 12. Classification results on the IP dataset. (a) Ground truth. (b) SVM (73.97%). (c) 3-D-CNN (76.25%). (d) SSRN (87.53%). (e) SF (81.54%). (f) MGHT (91.32%). (g) CEGCN (92.51%). (h) HGCN-MHF (93.99%). (i) AMGCFN (93.40%). (j) WFCG (91.74%). (k) GTFN (90.09%). (l) HGTFN (95.73%).

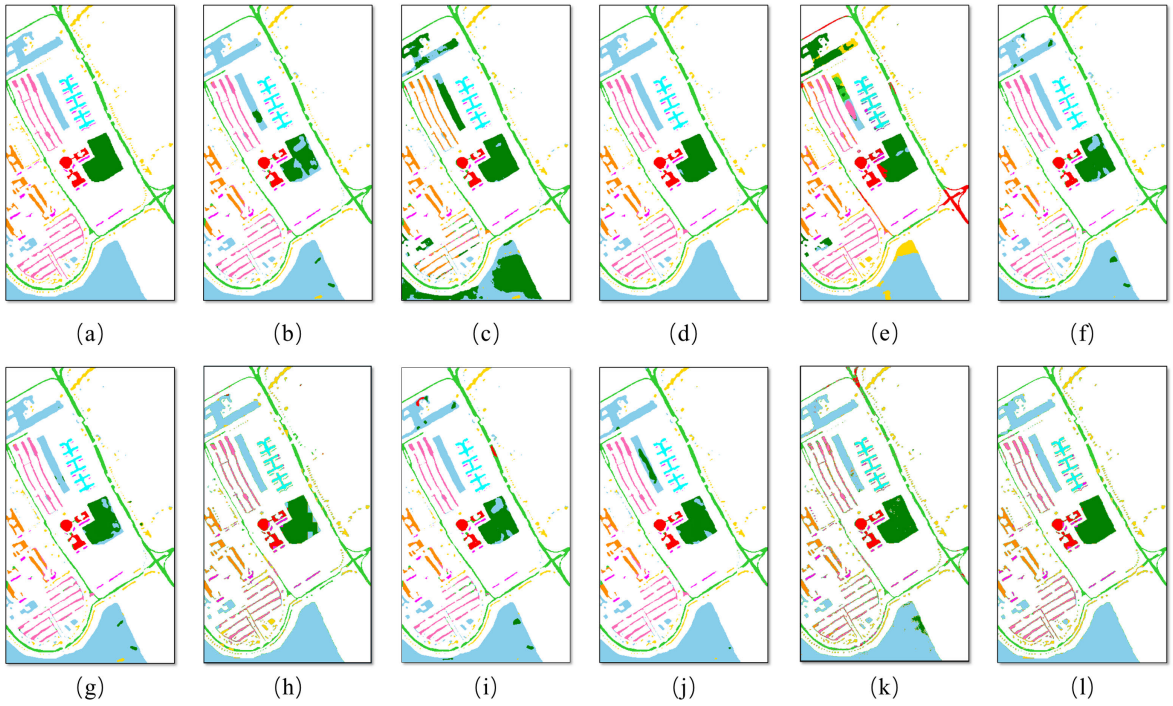


Fig. 13. Experimental results comparing different algorithms on the PU dataset. (a) Ground truth. (b) SVM (83.73%). (c) 3-D-CNN (82.29%). (d) SSRN (93.05%). (e) SF (78.17%). (f) MGHT (93.40%). (g) CEGCN (95.40%). (h) HGCN-MHF (94.16%). (i) AMGCFN (94.83%). (j) WFCG (94.97%). (k) GTFN (91.50%). (l) HGTFN (95.58%).

due to over-smoothing in hypergraph propagation or degraded attention quality. These findings suggest that shallow to moderately deep configurations strike a better balance between expressive capacity and optimization stability. Considering the tradeoff between classification performance and computational efficiency, the depth of HGCEA and KPFormer was fixed to 2 in all subsequent experiments.

D. Classification Results

This section compares the proposed HGTFN model with traditional machine learning and advanced deep learning methods.

1) Comparative Experiment Analysis: To comprehensively evaluate the performance of the proposed HGTFN, we conducted experiments on four benchmark hyperspectral datasets: IP, PU, SA, and HongHu. The results are summarized in Tables V–VIII, bolding is the best result. Across all datasets, HGTFN consistently achieves superior accuracy, demonstrating excellent generalization ability and structural modeling capability.

On the IP dataset, HGTFN achieves an OA of 95.73%, with particularly strong performance on categories 2, 5, 7, 8, 11, and 12. For example, in class 2, HGTFN achieves 92.13%, clearly outperforming MGHT (81.34%) and

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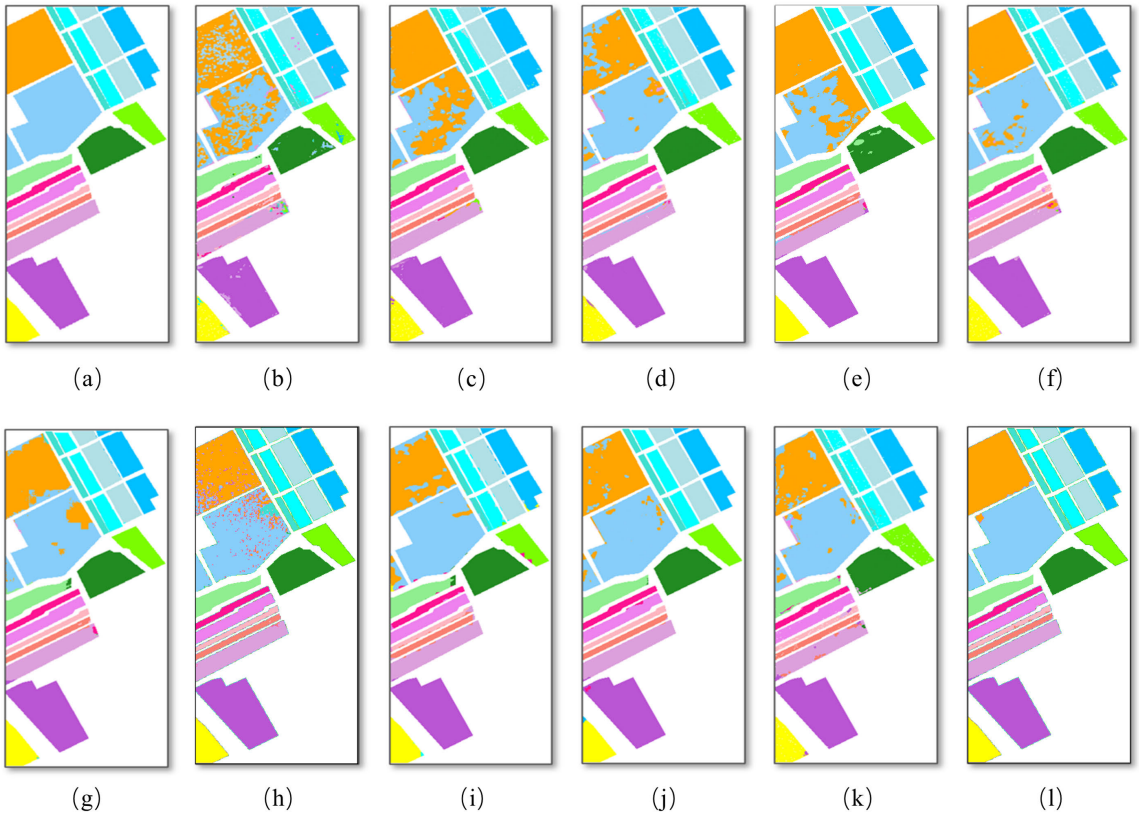


Fig. 14. Experimental results comparing different algorithms on the SA dataset. (a) Ground truth. (b) SVM (85.18%). (c) 3-D-CNN (88.26%). (d) SSRN (94.67%). (e) SF (91.13%). (f) MGHT (95.33%). (g) CEGCN (95.79%). (h) HGCN-MHF (94.71%). (i) AMGCFN (95.17%). (j) WFCG (95.13%). (k) GTFN (94.32%). (l) HGTFN (99.19%).

CEGCN (84.52%). Similarly, for class 7, HGTFN attains 100.00% accuracy, while most other methods remain below 93%. These results highlight HGTFN's ability to discriminate between complex terrain types and eliminate boundary ambiguities.

In the PU dataset, HGTFN showed consistent performance across all nine land cover categories. It has the highest accuracy of 98.66% and 96.49% on class 7 and class 8, respectively, which indicates its superiority in capturing the fine-grained spectral and textural variations of man-made materials. Meanwhile, the excellent performance of HGTFN on class 6 and class 2 further illustrates that HGTFN is able to effectively model vegetation structures. These validate its strength in modeling fragmented and spectrally complex urban areas. Such performance benefits from the proposed adaptive hyperedge weighting (ASSHW) and key-pixel semantic modeling, which jointly enhance robustness to intraclass spectral variations and spatial ambiguity.

On the SA dataset, HGTFN achieves a new state-of-the-art with an OA of 99.19%, AA of 98.85%, and Kappa of 99.10. It achieves 100.00% accuracy in 5 classes (1, 2, 3, 9, 11, 16), and performs significantly better than other models in several challenging classes. In class 8, HGTFN achieves 99.07%, while GTFN, WFCG, and AMGCFN only reach 85.26%, 88.36%, and 95.13%, respectively. Similarly, in class 15, HGTFN obtains 98.75%, outperforming HGCN-MHF (74.93%), GTFN (87.07%), and MGHT (92.86%). These results suggest that HGTFN can accurately capture fine-grained spectral variations and boundary details in densely cultivated regions, which are often overlooked by conventional methods. This gain is mainly attributed to the HGCEA module's ability to model complex boundary structures via high-order hyperedges, and the key-pixel-aware token construction that enhances class-specific discrimination.

On the HongHu dataset, HGTFN also achieves the highest OA of 96.56% and AA of 95.70%. It ranks first in 8 out of 22 classes, with particularly large margins in classes 5, 8, and 12. For instance, in class 8, HGTFN reaches 99.14%, outperforming the next-best method (HGCN-MHF) by over 3.4%, and other methods by more than 10%. In class 12, HGTFN achieves 97.52%, while most baselines remain below 90%, highlighting its ability to handle noisy and underrepresented categories. We attribute these improvements to the RCFN, which enhances feature integrity under challenging conditions such as noise, occlusion, or class imbalance.

2) Visual Analysis of Classification Results: As illustrated in Figs. 12–15, the classification maps on four representative hyperspectral datasets provide an intuitive comparison of different methods in terms of spatial boundary preservation, category consistency, and detail reconstruction.

For the IP dataset, HGTFN maintains high intraclass continuity across large agricultural areas. In contrast, other methods tend to produce blurred or fragmented regions around class borders. In the PU dataset, HGTFN effectively reduces

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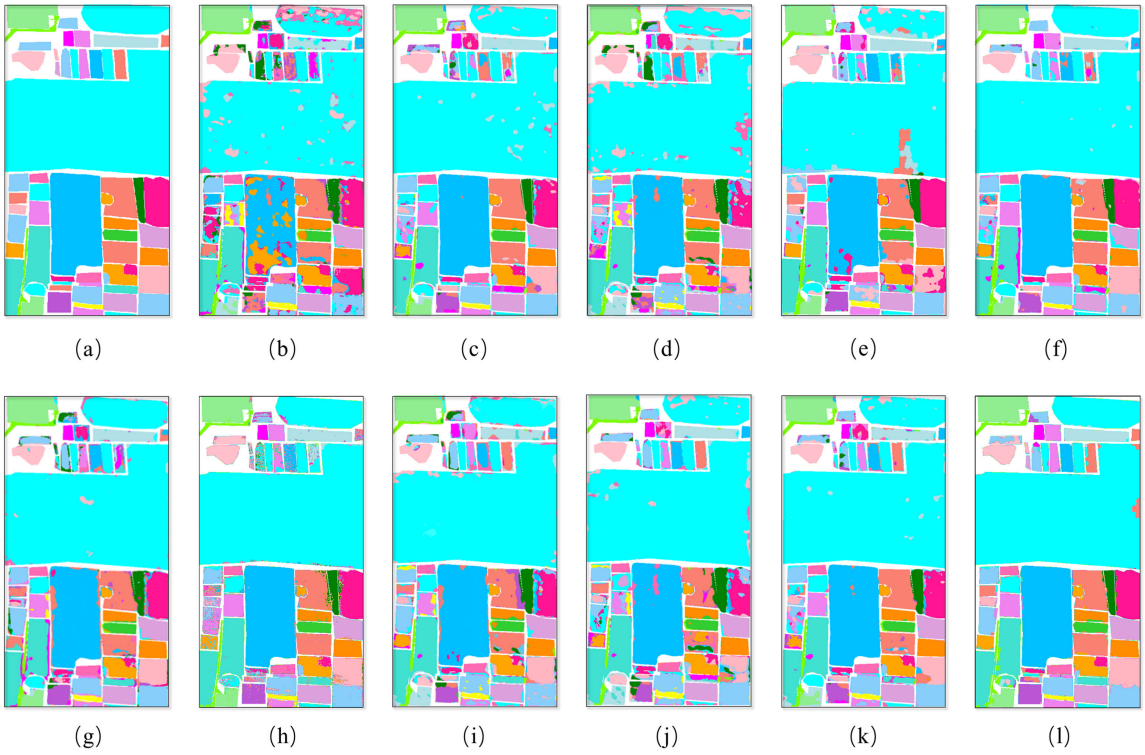


Fig. 15. Experimental results comparing different algorithms on the HongHu dataset. (a) Ground truth. (b) SVM (78.83%). (c) 3-D-CNN (87.69%). (d) SSRN (89.47%). (e) SF (89.13%). (f) MGHT (94.82%). (g) CEGCN (92.30%). (h) HGCN-MHF (92.63%). (i) AMGCFN (94.22%). (j) WFCG (92.54%). (k) GTFN (92.21%). (l) HGTFN (96.56%).

TABLE IX
RESULTS OF DIFFERENT CONFIGURATIONS FOR SUBNETWORK

Data	HGCEA	KPFormer	RCFN	OA(%)	AA(%)	κ
IP	×	√	×	90.70	94.61	89.28
	√	×	×	91.07	93.96	89.81
	√	√	×	92.71	94.90	91.65
	√	√	√	**94.98**	**96.26**	**94.20**
PU	×	√	×	92.50	87.67	90.07
	√	×	×	90.91	84.36	88.04
	√	√	×	93.83	90.65	91.97
	√	√	√	**95.42**	**92.38**	**93.92**
SA	×	√	×	95.11	96.40	94.54
	√	×	×	96.71	96.79	96.17
	√	√	×	98.37	98.61	97.51
	√	√	√	**99.39**	**99.12**	**99.32**
HongHu	×	√	×	86.06	86.41	82.76
	√	×	×	93.38	92.15	91.68
	√	√	×	94.57	94.34	93.34
	√	√	√	**95.54**	**94.06**	**93.96**

Note: "√" indicates the module is included; "×" indicates it is excluded. Each set of experiments was performed 10 times and averages were selected, with optimal results indicated in bold.

TABLE X
RESULTS OF DIFFERENT CONFIGURATIONS FOR ASSHW

e_{ELAH}	e_{NAH}	e_{FASH}	ASSHW	IP	PU	SA	HongHu
×	×	√	×	87.41	94.47	97.30	92.49
√	×	√	×	90.18	94.82	97.55	93.32
×	√	√	×	89.08	94.90	97.19	92.95
√	√	√	×	93.40	95.01	98.26	94.96
√	√	√	√	**95.43**	**95.94**	**99.35**	**96.07**

Note: "√" denotes that the corresponding hyperedge type or weighting strategy is included; "×" means excluded. Each set of experiments was performed 10 times and averages were selected, with optimal results indicated in bold.

confusion between spectrally similar urban structures, especially between green-colored roads and red rooftops, where it preserves distinct boundaries. For the SA, which contains numerous fine-grained crop regions, HGTFN exhibits remarkable accuracy and spatial smoothness, especially in elongated and fragmented plots that are prone to misclassification. Meanwhile, in the HongHu dataset, our method demonstrates robust discrimination between cotton bodies and vegetation. Notably, it achieves clearer segmentation boundaries and better preservation of small-scale categories, which are often neglected by existing methods.

Overall, the visual comparison across all four datasets demonstrates the consistent superiority of the proposed HGTFN in both spatial consistency and boundary delineation. These visual results validate the effectiveness of HGTFN in modeling complex spatial–spectral relationships, enabling both local detail preservation and global context integration. This advantage becomes more apparent in heterogeneous scenes and fragmented regions, highlighting its adaptability and generalization capacity across diverse hyperspectral datasets.

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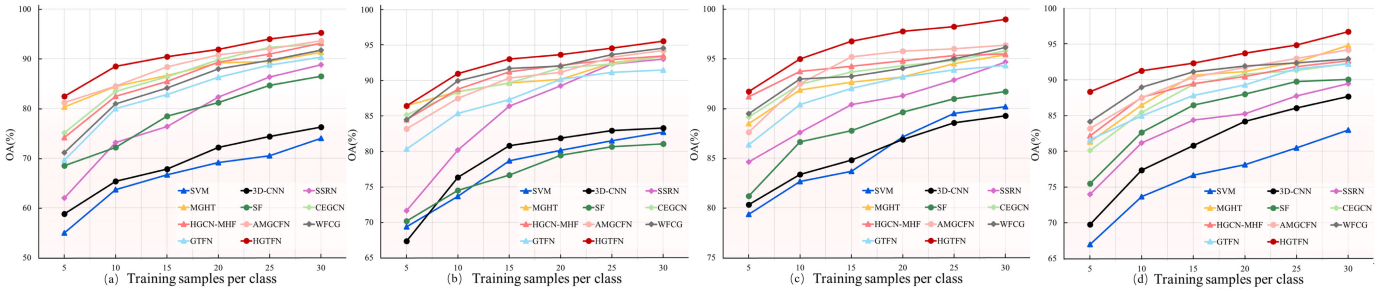


Fig. 16. OA for different proportions of training samples on the four datasets. (a)–(d) OA curves for the IP, SA, PU, and Honghu datasets, respectively.

TABLE XI
COMPARISON OF PARAMETERS, FLOPS, TRAINING AND TESTING TIME FOR DIFFERENT METHODS

Metrics	CNN-based Methods		Transformer-based Methods			Graph-based Methods			HGTFN
	3D-CNN	SSRN	MGHT	SF	GTFN	CEGCN	AMGCFN	WFCG	
Parameters	313964	540808	965760	604672	281878	166370	368217	102422	159969
FLOPs (G)	0.69	1.20	6.10	5.62	2.40	4.68	6.32	5.98	1.58
Training time (s)	87.96	108.42	162.59	468.67	136.98	53.69	89.61	76.75	127.58
Testing time (s)	4.82	3.56	2.98	41.92	7.06	1.96	0.96	1.24	0.23

V. DISCUSSION

A. Effectiveness of Subnetwork

To evaluate the individual and combined effects of the HGCEA, KPFormer, and RCFN modules, we conduct ablation experiments on all four datasets. The classification performance is summarized in Table IX. The experimental results show that, as a key-pixel orientated transformer encoder, KPFormer is able to capture long-distance dependencies between representative pixels. This structure performs well across all datasets, especially on spatially fragmented scenes like IP and PU, where it reaches OA values of 92.71% and 92.50%, respectively. However, due to the lack of structural priors, it shows relatively weaker performance on SA and HongHu, particularly in distinguishing fine-grained local variations. HGCEA leverages a hypergraph convolution enhanced by edge-wise attention, effectively modeling spatial–spectral structures. On the SA and HongHu datasets, which contain highly structured agricultural and wetland regions, HGCEA outperforms KPFormer alone by significant margins, indicating its strong ability to preserve object boundaries and capture contextual consistency. Combining HGCEA with KPFormer results in substantial performance gains on all datasets, confirming the complementary nature of local hypergraph modeling and global semantic encoding. Introducing RCFN on top of the dual-branch architecture further boosts performance, particularly in AA and κ, which reflects better intraclass consistency and stability in small-object classification. In summary, HGCEA enhances structural perception, KPFormer captures long-range semantics, and RCFN provides residual-based pixel refinement, with their synergy yielding a more accurate and robust HSI classifier.

B. Effectiveness of ASSHW

To comprehensively evaluate the individual and combined effects of the three proposed hyperedge components (e_{ELAH}, e_{NAH}, and e_{FASH}) and the adaptive weighting strategy, we conduct step-by-step ablation experiments on four benchmark datasets. The results are summarized in Table X.

It can be observed that incorporating e_{FASH} alone already achieves robust classification performance, particularly on the SA dataset, which is known for its high spectral redundancy. This highlights the superiority of e_{FASH} in modeling nonlocal spectral correlations. Building on this, the sequential introduction of e_{ELAH} and e_{NAH} further enhances the model's capability in capturing fine-grained boundary structures and preserving regional contextual consistency. When all three types of hyperedges are integrated, the classification performance improves consistently across all datasets, confirming the effectiveness of the proposed multiscale hypergraph construction strategy in modeling both local details and global dependencies. Furthermore, by introducing the ASSHW module on top of the fused hyperedge set, the model gains the ability to perform content-adaptive hyperedge weighting based on spatial and spectral similarities. This significantly strengthens the model's capacity to integrate complementary structural cues. The final model achieves the best performance on all datasets, with an OA of 95.43% on the IP dataset and a remarkable 99.35% on the SA dataset, demonstrating the crucial role of ASSHW in enhancing the robustness and generalization of the hypergraph structure.

C. Analysis of Generalization Performance

In order to verify the generalization performance of HGTFN, a small number of labeled samples from each category, i.e., (from 5 to 30), were selected to compare the classification results of the different methods. Fig. 16 shows the OA curves on the IP, SA, PU, and HongHu datasets. It can be seen that the proposed HGTFN obtains the highest OA on all training conditions and datasets, showing strong generalization capability and superior sample efficiency. Across all four datasets, HGTFN consistently outperforms other methods in most scenarios, particularly under limited training samples. On the SA dataset, its OA steadily exceeds 97% as the

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number of samples increases. In the large-scale HongHu dataset, HGTFN achieves the highest OA (88.53%) even with scarce data. Overall, HGTFN delivers consistent improvements over both traditional and graph-based approaches, validating the effectiveness of integrating hypergraph modeling with key-pixel-guided transformer encoding and highlighting its superiority in generalization performance.

D. Complexity Analysis

To evaluate the computational complexity and runtime efficiency of different methods, we compare the number of parameters, FLOPs, training time, and inference time, as summarized in Table XI. Overall, CNN-based models (e.g., 3-D-CNN and SSRN) exhibit relatively low parameter counts and FLOPs, resulting in faster inference. However, their limited receptive field restricts the capacity to capture complex dependencies. Transformer-based methods (e.g., MGHT, SF, and GTFN) leverage attention mechanisms to model global relationships, achieving stronger representational power at the cost of significantly increased FLOPs and training overhead. Graph-based approaches (e.g., CEGCN, AMGCFN, and WFCG) offer a more balanced tradeoff between complexity and efficiency. In comparison, the proposed HGTFN integrates hypergraph convolution and transformer modules to enhance global context modeling. In terms of complexity, HGTFN ranks second in parameter count and third in FLOPs, substantially lower than most transformer-based models, thus avoiding excessive attention-related overhead. Moreover, its parameter count and FLOPs remain comparable to or even better than several graph-based counterparts. In addition, HGTFN achieves the fastest inference time of 0.23 s, ranking first among all methods, which underscores its high efficiency. In summary, HGTFN not only achieves the best classification accuracy but also maintains a favorable balance in parameter count, computational cost, and inference speed—demonstrating strong potential and competitiveness for practical applications.

VI. Conclusion

In this article, we propose a novel dual-branch classification framework, named HGTFN, to address the limitations in high-order spectral–spatial dependency modeling and long-range contextual feature extraction in HSIC. The proposed framework integrates hypergraph convolution and transformer encoding to jointly exploit structured relationships and global semantics. On the hypergraph branch, diverse adaptive hyperedges and edge-wise attention mechanisms are introduced to capture fine-grained spectral–spatial relations. On the transformer branch, a key-pixel-guided tokenization strategy is designed to adaptively construct local sequences for salient regions, enabling efficient global context modeling. A residual complementary fusion network is further employed to dynamically integrate both branches, enhancing semantic consistency while improving discrimination in complex boundaries and heterogeneous regions. Meanwhile, the extensive experiments conducted in this article on four benchmark datasets demonstrate the superior performance of HGTFN. Under better model complexity and FLOPs, HGTFN significantly outperforms representative CNN, transformer, and graph-based models in terms of accuracy and generalization. Especially in the small-sample case, HGTFN maintains a clear advantage, demonstrating its strong potential for sample efficiency and cross-domain robustness.

In future work, we plan to design a lightweight transformer backbone and an adaptive key-pixel selection strategy to further reduce computational costs. In addition, we will explore a dynamic receptive field adjustment mechanism to improve adaptability to varying spatial structures, thereby paving the way for more efficient and scalable deployment of HGTFN in real-world applications. Inspired by recent advances in fuzzy HGNNs [54], we will further explore uncertainty-aware hypergraph modeling for HSIC. By incorporating fuzzy membership functions or belief-based weighting schemes into the hypergraph construction and convolution processes, we aim to effectively mitigate data heterogeneity and enhance model robustness in complex scenes.

References

[1] Q. Tong, B. Zhang, and L. Zhang, "Frontier progress in hyperspectral remote sensing in China," *J. Remote Sens.*, vol. 20, no. 5, pp. 689–707, 2016.

[2] H. Chen, J. Gan, and Q. Liu, "Application and prospect of hyperspectral remote sensing technology in geoenvironmental monitoring," *Urban Geol.*, vol. 19, no. 2, pp. 176–183, 2024.

[3] X. Zhang, "Application of remote sensing technology in agricultural production," *Hans J. Agricult. Sci.*, vol. 8, no. 7, pp. 776–780, 2018.

[4] Y. Zhang, Y. Guo, R. Tang, and Z. Tang, "Progress and trends in the application of hyperspectral remote sensing in plant diversity research," *J. Remote Sens.*, vol. 27, no. 11, pp. 2467–2483, 2023.

[5] Y. Ma, W. Zhang, and D. Liu, "Characterization of hyperspectral reconnaissance technology and analysis of its threat to military targets on the ground," *Shanghai Aerosp. Corp.*, vol. 29, no. 1, pp. 37–40, 2012.

[6] F. Ullah, I. Ullah, R. U. Khan, S. Khan, K. Khan, and G. Pau, "Conventional to deep ensemble methods for hyperspectral image classification: A comprehensive survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3878–3916, 2024.

[7] Y. Cai, Z. Zhang, P. Ghamisi, Z. Cai, X. Liu, and Y. Ding, "Fully linear graph convolutional networks for semi-supervised and unsupervised classification," *ACM Trans. Intell. Syst. Technol.*, vol. 14, no. 3, pp. 1–23, Jun. 2023.

[8] D. Al-Alimi, Z. Cai, M. A. A. Al-Qaness, E. Ahmed Alawamy, and A. Alalimi, "ETR: Enhancing transformation reduction for reducing dimensionality and classification complexity in hyperspectral images," *Expert Syst. Appl.*, vol. 213, Mar. 2023, Art. no. 118971.

[9] W. Cai, M. Gao, Y. Ding, X. Ning, X. Bai, and P. Qian, "Stereo attention cross-decoupling fusion-guided federated neural learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 552751.

[10] S. Sawant and M. Prabukumar, "A survey of band selection techniques for hyperspectral image classification," *J. Spectral Imag.*, vol. 9, no. 1, p. 5, Jun. 2020.

[11] Y. Ding et al., "Self-supervised locality preserving low-pass graph convolutional embedding for large-scale hyperspectral image clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536016.

[12] C. Bo, H. Lu, and D. Wang, "Spectral–spatial k-nearest neighbor approach for hyperspectral image classification," *Multimedia Tools Appl.*, vol. 77, no. 9, pp. 10419–10436, May 2018.

[13] S. Zhong, C.-I. Chang, and Y. Zhang, "Iterative support vector machine for hyperspectral image classification," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 3309–3312.

[14] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[15] Y. Xu, B. Du, F. Zhang, and L. Zhang, "Hyperspectral image classification via a random patches network," *ISPRS J. Photogramm. Remote Sens.*, vol. 142, pp. 344–357, Aug. 2018.

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[16] M. He, B. Li, and H. Chen, "Multi-scale 3D deep convolutional neural network for hyperspectral image classification," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 3904–3908.

[17] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[18] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[19] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, Jan. 2021.

[20] M. Ahmad, M. Mazzara, and S. Distefano, "Regularized CNN feature hierarchy for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 12, p. 2275, Jun. 2021.

[21] S. Wan, C. Gong, S. Pan, J. Yang, and J. Yang, "Multi-level graph convolutional network with automatic graph learning for hyperspectral image classification," 2020, *arXiv:2009.09196*.

[22] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 30–31.

[23] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[24] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[25] X. Zhang, Y. Su, L. Gao, L. Bruzzone, X. Gu, and Q. Tian, "A lightweight transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5517617.

[26] E. Ouyang, B. Li, W. Hu, G. Zhang, L. Zhao, and J. Wu, "When multigranularity meets spatial–spectral attention: A hybrid transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5559874.

[27] X. Zhao, J. Ma, L. Wang, Z. Zhang, Y. Ding, and X. Xiao, "A review of hyperspectral image classification based on graph neural networks," *Artif. Intell. Rev.*, vol. 58, no. 6, p. 172, Mar. 2025.

[28] F. F. Shahraki and S. Prasad, "Graph convolutional neural networks for hyperspectral data classification," in *Proc. IEEE Global Conf. Signal Inf. Process. (GlobalSIP)*, Nov. 2018, pp. 968–972.

[29] A. Qin, Z. Shang, J. Tian, Y. Wang, T. Zhang, and Y. Y. Tang, "Spectral–spatial graph convolutional networks for semisupervised hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 241–245, Feb. 2019.

[30] Z. Ma, Z. Jiang, and H. Zhang, "Hyperspectral image classification using feature fusion hypergraph convolution neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517314.

[31] S. Wan, C. Gong, P. Zhong, S. Pan, G. Li, and J. Yang, "Hyperspectral image classification with context-aware dynamic graph convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 597–612, Jan. 2021.

[32] Y. Ding, Y. Guo, Y. Chong, S. Pan, and J. Feng, "Global consistent graph convolutional network for hyperspectral image classification," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–16, 2021.

[33] Y. Ding, C. Li, F. Luo, and C. Zheng, "Hyperspectral image classification based on subgraph-dependent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5515315.

[34] W. Yu, S. Wan, G. Li, J. Yang, and C. Gong, "Hyperspectral image classification with contrastive graph convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503015.

[35] P. Yang, L. Tong, B. Qian, Z. Gao, J. Yu, and C. Xiao, "Hyperspectral image classification with spectral and spatial graph using inductive representation learning network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 791–800, 2021.

[36] A. Sha, B. Wang, X. Wu, and L. Zhang, "Semisupervised classification for hyperspectral images using graph attention networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 157–161, Jan. 2021.

[37] V. Zayats and M. Ostendorf, "Conversation modeling on Reddit using a graph-structured LSTM," *Trans. Assoc. Comput. Linguistics*, vol. 6, pp. 121–132, Dec. 2018.

[38] Y. Ding, M. Hou, Y. Ding, C.-H. Zheng, Q. Yan, and D.-S. Huang, "Exploring positional distributions of labeled superpixels within graph convolutional networks for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5535115.

[39] C. Shi, H. Wu, and L. Wang, "CEGAT: A CNN and enhanced-GAT based on key sample selection strategy for hyperspectral image classification," *Neural Netw.*, vol. 168, pp. 105–122, Nov. 2023.

[40] Y. Duan, F. Luo, M. Fu, Y. Niu, and X. Gong, "Classification via structure-preserved hypergraph convolution network for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507113.

[41] Y. Wang, Z. Xue, M. Jia, Z. Liu, and H. Su, "Hypergraph convolutional network with multiple hyperedges fusion for hyperspectral image classification under limited samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5526318.

[42] T.-H.-H. Chan, A. Louis, Z. G. Tang, and C. Zhang, "Spectral properties of hypergraph Laplacian and approximation algorithms," *J. ACM*, vol. 65, no. 3, pp. 1–48, Jun. 2018.

[43] Y. Gao, Y. Feng, S. Ji, and R. Ji, "HGNN+: General hypergraph neural networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 3181–3199, Mar. 2023.

[44] Z. Zhang, Y. Cai, W. Gong, P. Ghamisi, X. Liu, and R. Gloaguen, "Hypergraph convolutional subspace clustering with multihop aggregation for hyperspectral image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 676–686, 2022.

[45] Q. Wang, J. Huang, T. Shen, and Y. Gu, "EHGNN: Enhanced hypergraph neural network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[46] Y. Duan, A. Li, F. Luo, Z. Jing, Y. Niu, and Y. Li, "Clustering hypergraph U-Net with high-order topology-aware pooling and unpooling for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5520715.

[47] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[48] A. Ben Hamida, A. Benoit, P. Lambert, and C. Ben Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[49] X. Zhao et al., "MGHT: A multi-granularity hyperspectral image classification method based on hybrid transformer," *Spectrosc. Lett.*, vol. 58, no. 10, pp. 1–17, Nov. 2025.

[50] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel- and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2021.

[51] H. Zhou, F. Luo, H. Zhuang, Z. Weng, X. Gong, and Z. Lin, "Attention multihop graph and multiscale convolutional fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508614.

[52] Y. Dong, Q. Liu, B. Du, and L. Zhang, "Weighted feature fusion of convolutional neural network and graph attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 1559–1572, 2022.

[53] A. Yang, M. Li, Y. Ding, D. Hong, Y. Lv, and Y. He, "GTFN: GCN and transformer fusion network with spatial–spectral features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 6600115.

[54] T. Yin, W. Ding, H. Ju, J. Huang, and Y. Chen, "The fuzzy hypergraph neural network model based on sparse k-nearest neighborhood granule," *Appl. Soft Comput.*, vol. 170, Feb. 2025, Art. no. 112721.

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